園道

圍道

THE ART OF SURROUNDING

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https://github.com/10500awa/weidao

2026

Preface

What's Go? A game, of course. But one might ask: How is it done precisely? What can be done and what not? Is it possible to extend its concepts to other games?

圍道—*Wéi Dào —The art of surrounding* Tries to answer such questions by constructing a rigorous theory of the game of Go, followed by its generalization on and out of the board.

Part I: Origins and format. We begin the study of the game by its history. We finish this part with the rules of the game and setting the motivation for the foundation.

Part II: Foundation. Now that we know the game, we are ready to build it formally. The two-dimensional version of the game is constructed. The board, the pieces, the colors, and the relations between them are defined. Followed by the concepts of group, captures, and SuperKo. The legality of a play is formalized via a theorem. The SuperKo detection is handled by Zobrist Hashing. The whole structure is named Go, and we begin to generalize it. Later, we discuss the *life in the game of Go* also known as *unconditional life* by interpreting, in this theory, the results of David Benson; also, we develop the theory around the *conditional life* (Briefly, WIP). At the very end we discuss about two papers from Kim Jin Bai, in the second one of those there is a theorem that uses Benson results and we develop a counter-example of a Theorem.

Part III: Abstraction. We now study the board from outside, taking in consideration a little intuiton about the three dimensional Go, a concept about the quantity of liberties of a board naturally appears, such concept leads to a problem about liberties and how to control how the growth whenever the board gets big (in dimension). We also attack the one dimensional Go and finally we try to look beyond The game of Go in order to find if the structure we have made up for it can suit up other games.

Part IV: Infinite Go. WIP

I would like to thank Gusti Cabaña for being my first reader and giving me direction when I needed it most. Thanks also go to Diego Barrios and Eros Girardi for reviewing this work and providing invaluable feedback. Lastly, to Naza, for putting up with me throughout all these years and the years to come.

Contents

Part I Origins and format **1**

1.1 围棋—Wéi Qí—, history and rules . . . 1
1.1.1 History of Go in China . . . 1
1.1.2 History of Go Models . . . 3
1.1.3 Rules and formats . . . 4

Part II Foundation **8**

2.1 Go in two dimensions and two players. . . . 8
2.2 Toward Movement . . . 12
2.2.1 Move Function . . . 12
2.2.2 Rules of a Legal Move . . . 12
2.2.3 Legal Move Theorem . . . 14
2.2.4 Scoring . . . 15
2.3 Go in n Dimensions and with m Players . . . 16
2.3.1 Captures and Points for m Players . . . 17
2.3.2 Komi for Multiple Players . . . 17
2.4 Positional Superko Detection . . . 18
2.4.1 Zobrist Hashing . . . 19
2.5 The Structure of Time . . . 22
2.5.1 The Game as a Trajectory . . . 22
2.5.2 State Dynamics . . . 23
2.5.3 Anatomy of a Move . . . 23
2.6 Life in the Game of Go . . . 24
2.6.1 Benson's Theorem . . . 26
2.6.2 Seki . . . 29
2.6.3 Remark on Kim's Theorem 2 . . . 30

Part III Abstraction **32**

3.1 Comparing Adjacencies . . . 32
3.1.1 Go in Three Dimensions . . . 33
3.1.2 Comparing Average Adjacencies . . . 34
3.1.3 Removing Nodes, Yes. But How? . . . 35
3.1.4 Toroidal Board . . . 36
3.2 Go in One Dimension . . . 39
3.3 Common Origin . . . 39
3.4 Toward Surrounding Games . . . 42
3.5 Conclusion . . . 45

Part IV Infinite Go **46**

4.1 Infinite SuperKo . . . 48

I: Origins and format

合抱之木，
生於毫末；

九層之臺，
起於累土；

千里之行，
始於足下。

A tree as big as a man's embrace
grows from a tiny shoot;

A tower of nine storeys
begins with a heap of earth;

The journey of a thousand li
starts from where one stands.

道德經—Dàodé jīng, Chapter 64

1.1 围棋—Wéi Qí—, history and rules

1.1.1 History of Go in China

The term 围棋 (Wéi Qí) originates from the characters 围 (Wéi), meaning 'to surround', and 棋 (Qí), which refers to a 'board game'. In the West, it is commonly known as Go, a name derived from the Japanese 囲碁 (Igo). Although its exact chronology remains uncertain, ancient literary, pictorial, and archaeological records demonstrate that the game was an integral part of Chinese culture since BCE times.



Figure 1.1 – A game of Go on a traditional board.

Regarding its origins, mythical tradition holds that the legendary Emperor Yáo (堯) (2356–2257 BCE) invented it to enlighten the mind of his son, Dān Zhū (丹朱). Another account claims that Emperor Shùn (舜) (2255–2208 BCE) created it to educate his son, Shāng Jūn (商均). Furthermore, several sources attribute its invention to Wǔ, a vassal of Emperor Xià Jié (夏桀) (1818–1767 BCE), a courtier also celebrated for having devised the card game [9].

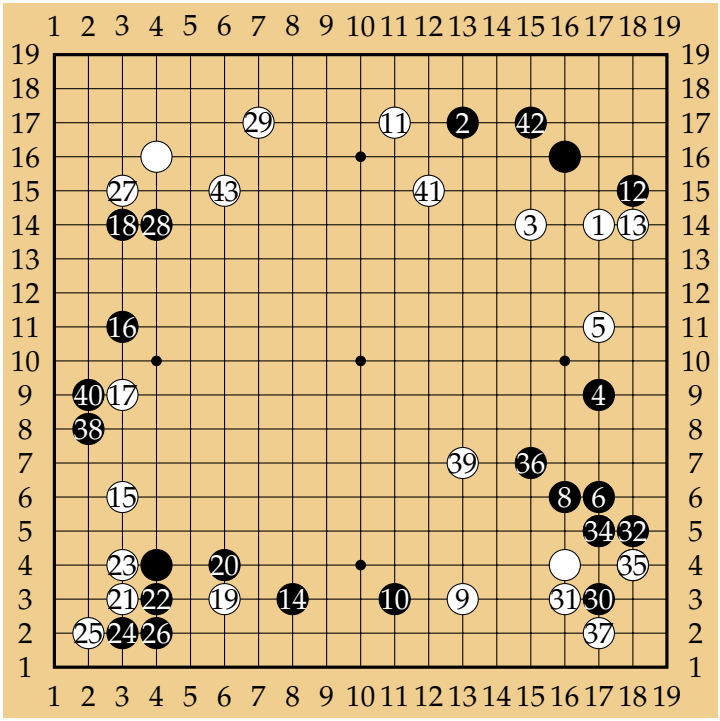


Figure 1.2 – Game between *Sūn Cè* and *Lǚ Fàn*. The unnumbered stones are due to the fact that, at that time, games began from a predetermined starting position and White played first.

The first reliable mention of the game dates to the 3rd century BCE, when Mèngzǐ (孟軻) describes Yì Qiū (弈秋), a master of the game during the Warring States period who was recognized as one of the foremost exponents of his time. On the other hand, a notable game is preserved in the Wàngyōu Qīnglè Jí (忘憂清樂集), a mid-12th-century compilation. This work records one of the earliest recorded games, dating from the Han period (2nd century BCE –2nd century CE) and contested between General Sūn Cè (孙策) and his captain Lǚ Fàn (Figure 1.2) [2]. In conclusion, while it is unlikely that Go is 4,000 years old—as certain traditions suggest—its existence has been documented for at least 2,500 years. Its recurring appearance in texts and chronicles of the period suggests that, by then, the game enjoyed widespread enough familiarity for analogies based on it to be readily understood.

1.1.2 History of Go Models

The vast literature on Go typically focuses on its complexity or on the search for optimal strategies, generally approaching them through *combinatorial game theory*. The present work belongs to the few studies devoted to *modeling* Go. This section is dedicated to tracing the historical development of such models.

The first paper exploring a potential mathematical model of Go was written by *Edward O. Thorp* and *William E. Walden* in 1964 [10]. In this article, the board is generalized from a fixed size to an $n \times m$ grid. Definitions of *adjacency*, the three states of any given intersection, and a basic formalization of the rules are established. Furthermore, they define Computer Go—a finite, cycle-free game—in order to study optimal strategies on small boards. They also introduce *t-truncated Go*, a variant that is forcibly terminated in a draw at turn t (if it has not ended earlier).

In 1970, *Albert L. Zobrist* published a method to encode a board position into a single large integer for state-verification purposes [12]. Although his work does not address Go per se, the method he describes will be employed in Section 2.4.1 to detect when one of the game's core rules, the *Superko* rule, is violated.

In 1972, Thorp and Walden published a combinatorial study exploring the nature and rules of Go to generate a computational model.

In 1976, David B. Benson [1] built upon Thorp and Walden's work to formally determine when a group is "alive". Although at first glance this contribution might seem separate from general game models, it stands as a pioneering attempt to model the internal dynamics of Go. His approach will be revisited under our own theory in Section 2.6.

Finally, mention must be made of the work of *Jin Bai Kim*, conceived in 1976 and published in two parts in 1978 [4, 5]. This study continues the tradition of using graphs to model Go, although Kim omits one of the possible board states and proposes an alternative definition of *adjacencies*. In the second of these articles, we encounter the first formulation of three-dimensional Go (which we shall examine in Part III).

1.1.3 Rules and formats

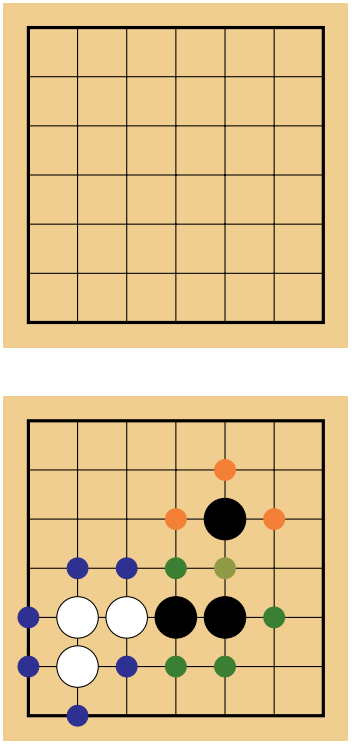


Figure 1.3 – An empty board, alongside a board illustrating the liberties of a single stone and of groups.

Go is played on a square board consisting of a grid of intersecting lines. It is contested by two players: one controls the black pieces and the other, the white ones. The most standard board sizes are 9×9, 13×13, and 19×19 (read as "nine-by-nine," "thirteen-by-thirteen," etc.). To introduce the rules, we shall employ a 7×7 board, a size more than adequate for this purpose.

As illustrated in Figure 1.2, coordinates are read starting from the lower-left corner, point $(1,1)$, with coordinates increasing to the right and upward. The game begins with an empty board (Figure 1.3), unless played with a handicap. The small black dots visible at certain intersections serve merely as visual reference points and provide standard placements for handicap stones.

On their turn, a player may perform one of two actions: place a piece (termed a *stone*) on an unoccupied intersection, or pass. The player controlling the black stones plays first and may place their stone on any vacant intersection of their choosing.

Intersections on a board have associated *adjacencies*. These correspond to other intersections located at a distance of one unit in either the horizontal or vertical direction. Upon placing a stone on the board, it acquires *liberties*, defined as the unoccupied intersections that are *adjacent* to the stone or to a *group*. Two or more adjacent stones of the same color constitute a *group*; groups share the *liberties* of each individual stone comprising them (see Figure 1.3).

It can be observed that, on the second board, the black stone has four liberties, the white group possesses six liberties, and the black group has five. Throughout the game, a stone or group may be captured; this is accomplished by *completely surrounding* it, thereby removing all the *liberties* it possesses. Figure 1.4 illustrates the capture of both a single stone and a group. When a capture occurs, the captured stones are removed from the board.

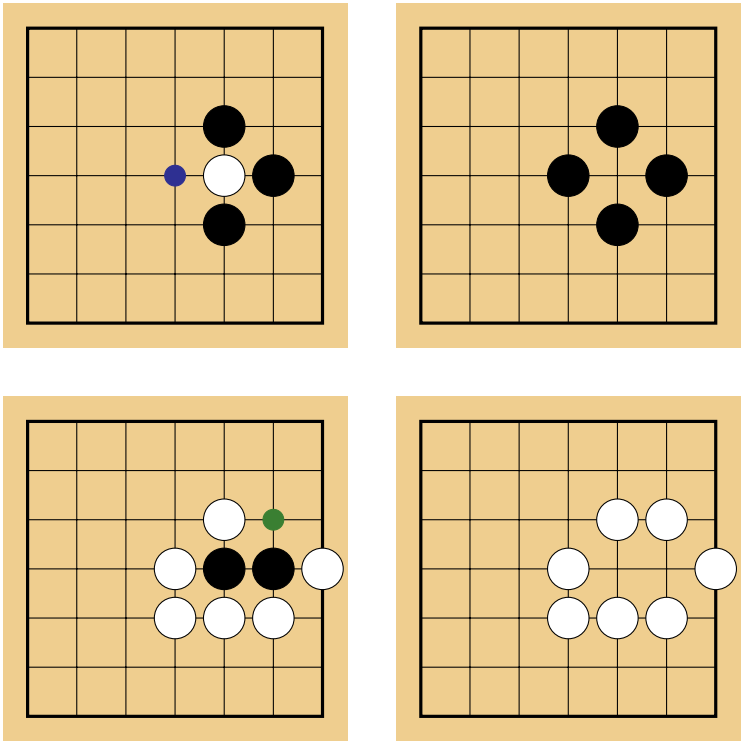


Figure 1.4 – Capture of a white stone and a black group.

For a stone to be placed on the board, it must have liberties at the intersection where it is played. The *suicide rule* prohibits placing stones on intersections that offer no liberties. Playing on such intersections would entail the immediate capture of the newly placed stone, tantamount to playing against oneself. Likewise, if a friendly group has only a single remaining liberty, filling that last vital space also constitutes suicide and is therefore an illegal move (see Figure 1.5).

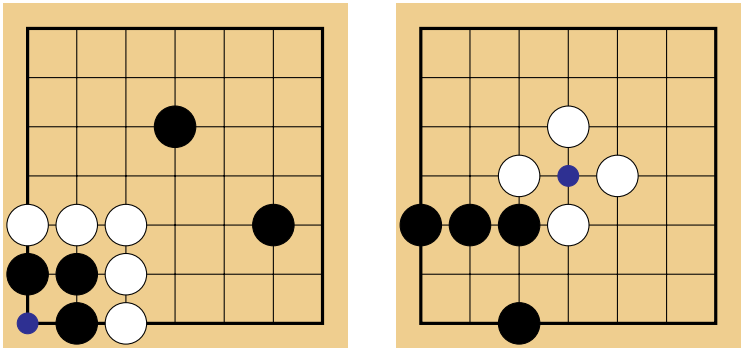


Figure 1.5 – In blue the illegal moves for Black as they are suicide.

There is, however, an exception to this rule: playing on an intersection with no liberties is legal if the move results in a capture. Upon removing the captured stones, the newly placed stone immediately gains liberties (see Figure 1.6).. Furthermore, as seen in the figure, if both players were allowed to play on that intersection, the cycle of reciprocal captures could continue indefinitely. To prevent this loop, the *ko rule* is introduced. If Black were to play at coordinates $(4, 4)$ (marked with a blue dot in the figure), the board would revert to the exact position of the preceding turn. The *ko rule*

forbids this.[1]

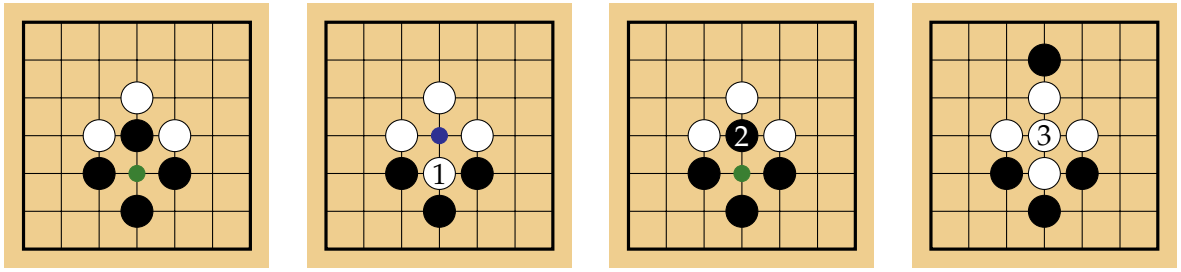


Figure 1.6 – An illegal position, as it reverts to a previous state. Following Black's move at $(4, 6)$, Black could recapture at $(4, 4)$ on a subsequent turn. White prevents this possibility by playing at $(4, 4)$.

To recapture, Black must first play elsewhere on the board (a ko threat). This generates a different board configuration; if the opponent responds, Black may then play at $(4, 4)$ to recapture, as the board state will no longer be identical to the earlier one (see the final board in Figure 1.6).

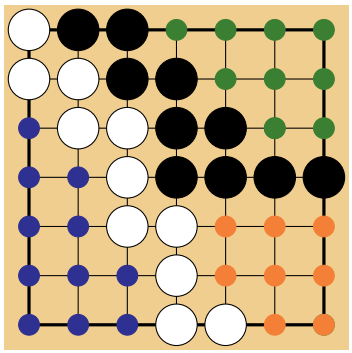


Figure 1.7 – Black territory, white territory and neutral.

An empty intersection (or a group thereof) is considered the *territory* of a given color if it is completely surrounded by stones of that color. Conversely, if an empty space is bounded by stones of both players, it is regarded as neutral, belonging to neither player (see Figure 1.7).

When two players of different skill levels face each other, a handicap is usually granted to the less experienced player. This advantage consists of playing with the black stones (moving first) and placing a predetermined number of *handicap stones* on the board before the game begins; the greater the skill disparity, the more stones are placed (see Figure 1.8).

This is the traditional placement of handicap stones (on both $13{\times}13$ and $19{\times}19$ boards). There also exists *free handicap placement*, in which the weaker player freely chooses where to position these initial stones.

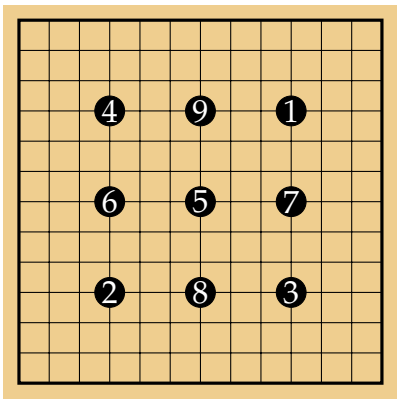


Figure 1.8 – Handicap of the 13×13 board.

The objective of Go is to secure more *territory* than the opponent. On their turn, a player may choose to pass, conceding the initiative to the rival. The game ends when both players pass consecutively. At that point, a phase of mutual agreement begins, during which all "dead" stones (that is, those whose eventual capture cannot be prevented) are identified and removed from the board. After clearing these stones, the winner is determined by summing the secured *territories* and the living stones remaining on the board at the end of the game. Figure 1.9 illustrates a completed game. The score would be computed as follows: 11 stones for Black plus 16 intersections in Black's territory; White,

[1] We will see that the standard *ko rule* proves insufficient in certain scenarios, as longer cycles can occur that repeat a previous board position without violating this basic rule (see Section 2.2.1).

for their part, has 11 stones and 11 intersections in territory. Consequently, Black scores 27 points and White 22, so Black wins by 5 points.[2]

One might oppose Black's victory, given the inherent advantage of making the first move. To offset this disparity, White is awarded a compensation for playing second, known as *komi*. This value varies according to board size and the skill difference between the players; generally, 5.5 points are used on standard boards.[3] Were this compensation applied to the preceding example, White would win by 0.5 points, the narrowest possible margin. The inclusion of this half-point in the *komi* precludes the possibility of a draw.

A group possesses an *eye* if it completely surrounds an empty intersection. With at least two *eyes*, it achieves the state of being *alive* and can never be captured. In Figure 1.10 (left), even if White attempts to attack the black group, they can only place one stone per turn. Because occupying either space would violate the suicide rule, White would need to play into both eyes simultaneously to capture, which is impossible. Consequently, the black group is *alive*.

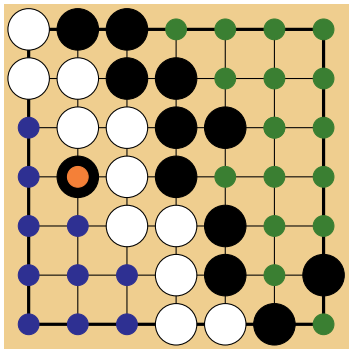


Figure 1.9 – Final board position of a completed game, with its intersections scored; the black stone marked with the orange dot must be removed as they are dead.

There exist, however, so-called false eyes: intersections that resemble *eyes*, but where one of the surrounding groups is at risk of being reduced to a single liberty (the eye). As a result, if the opponent succeeds in capturing this vulnerable group, the structure collapses and the eye is lost (see Figure 1.10, right).

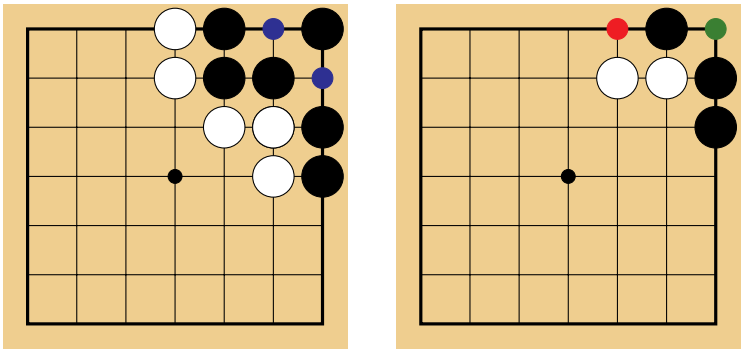


Figure 1.10 – Eyes of a living group (blue). False eye (green). If White plays at the red point, Black will be forced to fill their own eye with a stone, or risk losing it.

To build Go, we require four elements: *where*, *who*, *what*, and *how*.

[2] This method corresponds to Chinese scoring rules. There are several other counting systems, such as the Japanese or Korean rulesets.

[3] The exact value of *komi* depends on the chosen ruleset. Furthermore, in handicap games, a minimal *komi* of 0.5 points is typically applied.

II: Foundation

三十輻共一轂，
當其無，有車之用。

埏埴以為器，
當其無，有器之用。

鑿戶牖以為室，
當其無，有室之用。

故有之以為利，
無之以為用。

Thirty spokes converge to form a wheel hub.
But it is the emptiness inside the hub that makes it useful.

Clay is kneaded to form a pot.
But it is the emptiness inside it that makes it useful.

Doorways and window bays are chiselled to form a house.
But it is the emptiness inside it that makes it useful.

Substance brings benefits.
Emptiness brings usefulness.

道德經—Dàodé Jīng, Chapter 11

2.1 Go in two dimensions and two players.

Let us interpret the board in a convenient manner by considering just its points of intersection. Let $n, m \in \mathbb{N}$. We define $N = \{1, \ldots, n\}$ and $M = \{1, \ldots, m\}$. With these sets, we formalize the game space:[1]

$$G \coloneqq N \times M = \{(x, y) : x \in N,\ y \in M\}.$$

Let $t_1, t_2 \in G$ be two *points*[2]. If $t_1 = (a, b)$, we say that t_2 is adjacent to t_1, denoted by $t_1 \sim t_2$, if it satisfies exactly one of the following conditions:

$$t_2 = (a+1, b), \quad t_2 = (a, b+1), \quad t_2 = (a-1, b), \quad t_2 = (a, b-1).$$

If none of these conditions hold, t_1 is not adjacent to t_2, which we denote by $t_1 \nsim t_2$. Figure 2.1 illustrates the adjacencies of a point. Having established the *where*, let us examine *who* participates in it. When reviewing the rules, we observed that each point on the board can be in one of three possible states: unoccupied, occupied by a white stone, or occupied by a black stone. We define E as the set of states that a point may assume, $E = \{0, 1, 2\}$.

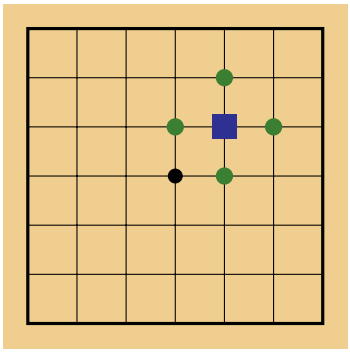


Figure 2.1 – Point $(5, 5)$ in blue, adjacencies in green.

These values represent, respectively: vacant, black, and white. With this, we can account for all possible

[1] We use the letter G because we will adopt Japanese Go terminology for our notation whenever it does not lead to confusion. In Japanese, the board is called *Goban* (碁盤).

[2] The letter t originates from the term *ten* (点), meaning point in Japanese (and Chinese).

states on the board:

$$E \times G = \{(e, t) : e \in E, t \in G\}.$$

For example, the point $(4, 4) \in G$ may be found in one of three states: $(0, (4, 4))$, meaning no stone at $(4, 4)$; $(1, (4, 4))$, a black stone at $(4, 4)$; or $(2, (4, 4))$, a white stone at $(4, 4)$.

We shall refer to the elements of $E \times G$ as *stones*, denoted by the letter i, such that $i = (e, t)$,[3] with $e \in E$ and $t \in G$. In Go, however, a single point cannot exhibit all states simultaneously; it assumes at most one.

Definition 2.1.1 (Board)**.** A subset $T \subseteq E \times G$ is said to be a *board* if,

$$\forall t \in G, \exists! e \in E \text{ such that } (e, t) \in T$$

That is, in T there is a unique state for each position. Having established the *where* and *who*, we can construct the *what*. To do so, we require two fundamental functions:

Definition 2.1.2 (Stone State Function)**.**

$$\text{We define } C : E \times G \to E, \quad (e, t) \mapsto e.$$

Definition 2.1.3 (Stone Position Function)**.**

$$\text{We define } P : E \times G \to G, \quad (e, t) \mapsto t.$$

Equipped with these, the notion of a group arises naturally:

Definition 2.1.4 (Connection)**.** Given a board T and $i_1, i_2 \in T$ two stones. We say that i_1 and i_2 are connected, denoted by $i_1 \leftrightarrow i_2$, if

$$C(i_1) = C(i_2) \quad \text{and} \quad P(i_1) \sim P(i_2).$$

Definition 2.1.5 (Group of Stones)**.** Let $T_c = \{i \in T \mid \exists i' \in T \text{ such that } i \leftrightarrow i'\}$ be the set of connected stones on a board T.

Two stones $i_a, i_b \in T_c$ belong to the same group (denoted by $i_a \equiv i_b$) if there exists a finite sequence of stones $\{i_1, i_2, \ldots, i_k\}$ $(k \geq 2)$ such that

$$i_1 = i_a, i_k = i_b \text{ and } i_m \leftrightarrow i_{m+1} \text{ for all } 1 \leq m < k.$$

The relation $\equiv$ is an equivalence relation on T_c. Its equivalence classes $I \in T_c/\equiv$ (each necessarily satisfying $|I| \geq 2$) are called groups, and the remaining elements $i \in T \setminus T_c$ are called stones.

Note on notation. Since isolated stones and groups share the same dynamic behavior regarding adjacencies, liberties, and captures, we shall denote stones (treated as singletons $\{i\}$) or groups by I^e, where e designates the state or color of the group; alternatively, we may write "*Let I be a group of color e*," or simply I whenever the color is not relevant.[4]

[3] From the term *ishi* (石), meaning "stone" in Japanese.

[4] From *ishi* (石, stone/stones). Denoted similarly to individual stones, but capitalized to signify a set thereof.

Definition 2.1.6 (Adjacencies of a Group)**.** Let I be a group of color e, and let $i = (e', t)$ be a stone on the board. We say that I is adjacent to i, denoted by $I \sim i$, if at least one stone in the group is adjacent to i:

$$I \sim i \iff \exists\, i_k \in I : P(i_k) \sim P(i).$$

These definitions likewise encompass the groups I^0. Given their fundamental importance, these groups are given a dedicated name: we shall refer to them as *regions* and denote them simply by the letter R.

To refer to the color of a group as a whole, we define $C(I^e) = e$, as every stone in I has the same color.

It will prove very useful to identify the adjacencies of a stone or group. To this end, we define the set K[5] associated with a stone or group (see Figure 2.2).

Definition 2.1.7 (Adjacency Set)**.** Given a board T, if i is a stone and I a group of stones of color e. We define the adjacency set of the stone as the stones satisfying the definition of adjacency, namely:

$$K(i) = \{i_r \in T : P(i) \sim P(i_r)\}.$$

For a group I, we define:

$$K(I) = \{i \in T \setminus I : I \sim i\}.$$

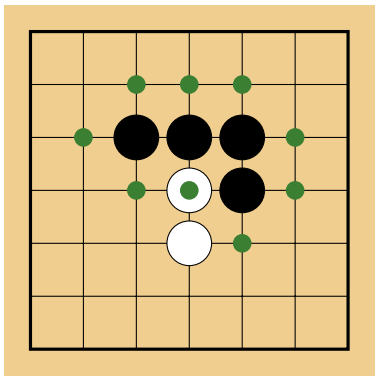


Figure 2.2 – A black group with its adjacencies marked.

Note that any adjacent stone of the same color already belongs to the group. Thus, if the group is of color e, the states of its adjacencies will belong, at most, to $E \setminus \{e\}$.

Abuse of Notation: Whenever a definition or theorem applies identically to both a group I and an isolated stone i, we shall let I denote either a group or an isolated stone, unless explicitly distinguished.

To characterize the states of these adjacencies, we will make use (and abuse) of the functions previously defined.

Definition 2.1.8 (State of the Adjacencies of a Group or Stone)**.** Given a board T, if I is a group of color e. Applying the function C elementwise to its adjacency set $K(I)$, we obtain the intended set, defined as:

$$C(K(I)) = \{C(i) \mid i \in K(I)\} \subseteq \{0, e'\}.$$

This definition is essential for describing much of the internal dynamics of Go. From now on, whenever we refer to groups I, it is to be understood that their color is distinct from 0. Whenever we deal with I^0, we shall employ the notation R, unless explicitly stated otherwise.

Definition 2.1.9 (Territory)**.** A region R is territory belonging to the player of color e if there exists a set S of groups of color e such that,

$$K(R) \subseteq \bigcup_{I \in S} I.$$

[5] From the term *chikaku* (近く, near/nearby).

That is to say, all stones adjacent to the region belong to groups of the same color. We shall denote the *territories*[6] of each color as:

$$J^e = \{R : C(K(R)) = \{e\}\}, \quad e \neq 0.$$

Regions that do not constitute territory for either player will be denoted and defined as

$$J^0 := \{R : |C(K(R))| \geq 2\}.$$

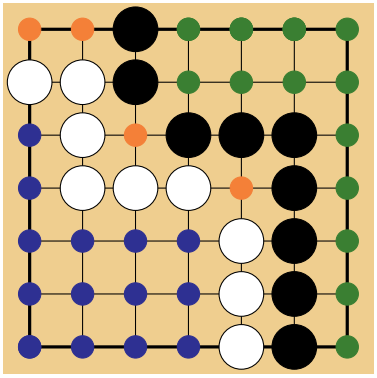


Figure 2.3 – White, black, and neutral territory.

Figure 2.3 illustrates the different types of *territories*. With this, we have established a partition of the set of *regions* (I^0), since every region belongs to White's territory, belongs to Black's territory, or forms part of neither player's territory:

$$I^0 = J^0 \cup J^1 \cup J^2.$$

Additionally, the cardinality of the adjacency set of an individual stone, $|K(i)|$[7], proves remarkably useful for determining its spatial placement on the board. Because $|K(i)|$ is bounded by Lemma 2.3.1, a straightforward analysis of the number of adjacencies reveals precisely whether the stone is situated in the interior, on an edge, or in a corner. This geometric property will be essential, and we shall make use of it in Sections 3.2 and 3.4.

To formalize captures, we first introduce the concept of a *liberty*.

Definition 2.1.10 (Liberties of a Group)**.** Let I be a group. We define the set of liberties of said group as:

$$L(I) = \{i \in K(I) : C(i) = 0\}.$$

Note that if $L(I) = \varnothing$ for a given group, then that group is considered *dead*. This situation can only occur upon executing a move and effecting a capture. But what constitutes a capture? What is a move?

A group is captured when it loses all of its liberties. Captured stones are removed from the board[8].

Definition 2.1.11 (Capture Criterion)**.** Let I be a group; then,

$$I \text{ is captured} \iff L(I) = \varnothing.$$

To formalize the removal of these stones from the board, if $I = \{i_1, i_2, \ldots, i_n\}$, it suffices to reassign their states by setting $C(i_j) = 0$ for all $j = 1, \ldots, n$.

We favor the phrasing *is captured*, since death is the consequence of a capture.

[6] From the term *ji* (地), meaning "land" in Japanese.
[7] This could be interpreted as the degree of a vertex in graph theory.
[8] We'll employ Chinese rules for scoring.

2.2 Toward Movement

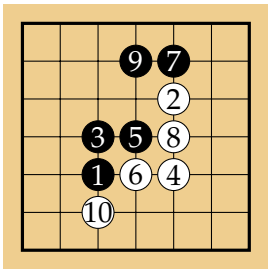
Thus far, we have described the static structure of the board. To model a game, we must incorporate the temporal evolution of play. To this end, we first define the space of all possible moves, which we denote by $\mathcal{J}$:

$$\mathcal{J} \coloneqq ((E \setminus \{0\}) \times G) \cup \{\varnothing\}.$$

2.2.1 Move Function

When playing Go on our turn, we either take a stone and place it on the board, or else we pass. We define the move function as $f : \mathbb{N} \to \mathcal{J}$. Let us examine an example.

$$\begin{aligned} f(1) &= (1,(2,2)), & f(2) &= (1,(4,4)), & f(3) &= (1,(2,3)), \\ f(4) &= (2,(4,2)), & f(5) &= (2,(3,3)), & f(6) &= (2,(3,2)), \\ f(7) &= (1,(4,5)), & f(8) &= (1,(3,4)), & f(9) &= (1,(3,5)), \\ f(10) &= (2,(2,1)), & f(11) &= \varnothing, & f(12) &= \varnothing. \end{aligned}$$



2.2.2 Rules of a Legal Move

The function records the moves, but not their legality. Lets analyze the rules. The *suicide rule* prohibits placing a stone if doing so results in the immediate capture of one's own group (Figure 2.4). There exists a single exception: a move played with no liberties is legal if it simultaneously captures a group of the opposite color (see Figure 2.5). Note that, due to this mechanic, when it is Black's turn, only white groups may be captured, and vice versa.

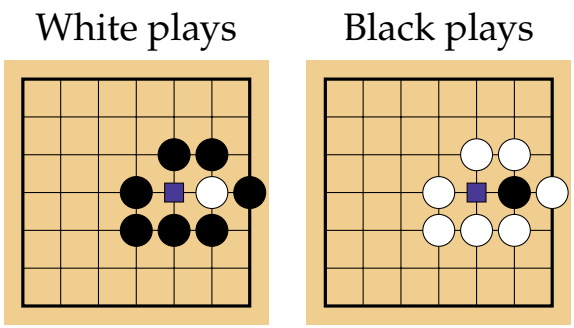


Figure 2.4 – In purple, illegal moves, as they lead to suicide.

To evaluate a move, we first verify whether it results in a capture. If so, the move is legal. If it does not capture, we inspect its liberties: if it has none, it constitutes suicide and is therefore illegal; otherwise, it is legal. As noted in Section 1.1.3, we must keep the *ko rule* in mind. This rule forbids the immediate recapture of a single stone if doing so reverts the board state precisely to the position of the preceding turn (see Figure 2.6).

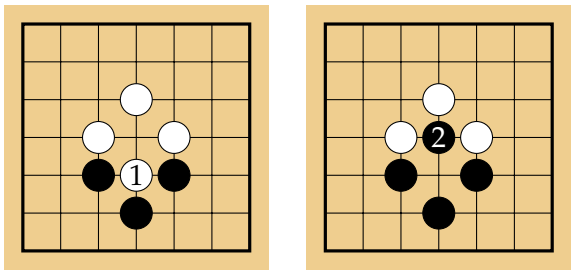


Figure 2.5 – Position of a stone under apparent suicide that is legal because it captures.

The basic *ko rule* is insufficient, as cycles exist where the board position is repeated without immediately recapturing the same stone. See Figure 2.7, in which an earlier position is reached once again via a sequence of captures that do not violate the basic *ko rule*.

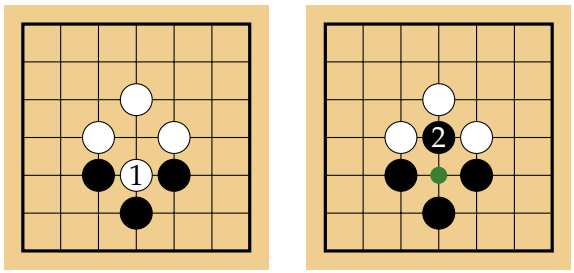


Figure 2.6 – White cannot play at the green intersection, as doing so repeats a prior board position.

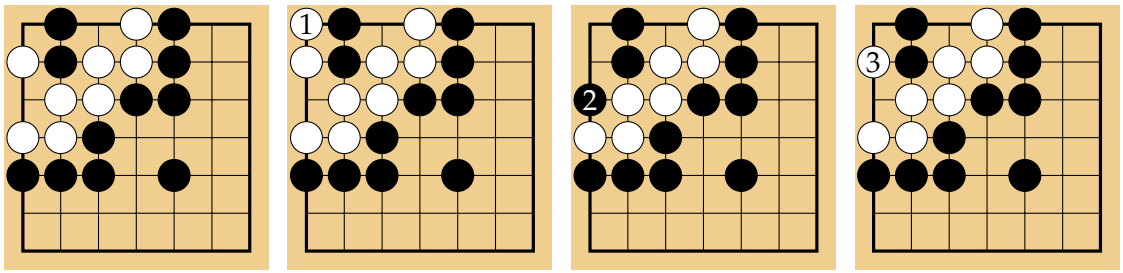


Figure 2.7 – Board configuration featuring a three-move cycle.

Because repeating positions is prohibited, we require a more general rule. To illustrate an extreme case, see Figure 2.8.[9]

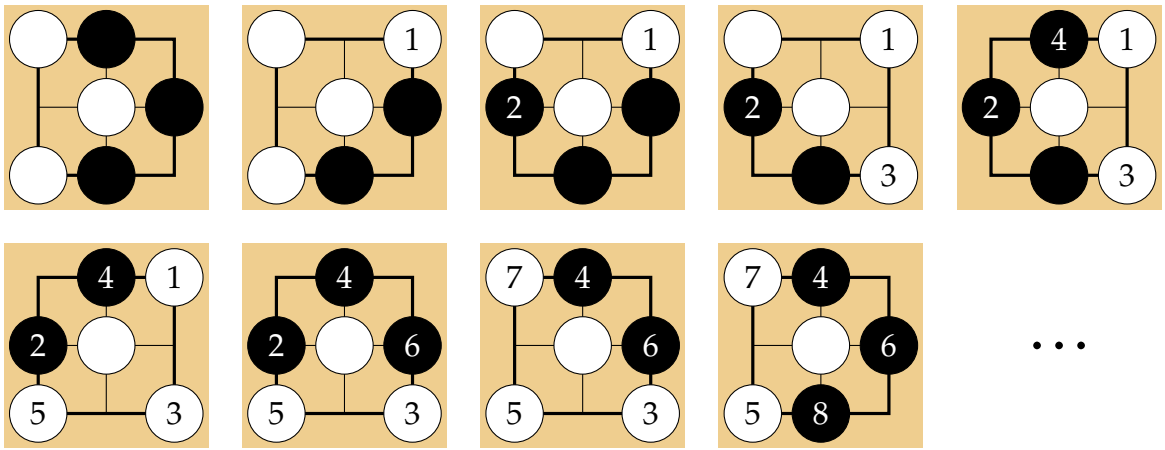


Figure 2.8 – A cycle of length 8 that can be repeated indefinitely in the absence of a Superko rule.

One might argue that on the board in Figure 2.7, the position has indeed repeated, but the turn belongs to the other player; hence, it is not a full repetition. This line of reasoning leads to what is known as the *Natural Superko* rule. We, however, shall adopt a more restrictive variant:

Definition 2.2.1 (Positional Superko)**.** A player may not place a stone in such a way that it recreates a board position that has already appeared previously in the game.

[9]There exist examples involving far longer cycles; to explore them in greater depth, see [8].

By this rule, in both Figure 2.7 and Figure 2.8, any attempt to repeat a cycle will require intermediate moves that generate genuine game progress.

It remains to consider the most elementary restriction in Go: it is not permitted to play on an intersection that is already occupied by another stone.

Definition 2.2.2 (Vacant State Condition)**.** For a move $f(n) = i$ to be registered at turn n, the corresponding point must previously be unoccupied. This is expressed as:

$$f(n) = i \text{ is playable} \iff C(i) = 0.$$

Having established the *where*, *who*, and *what*, let us examine the *how*.

2.2.3 Legal Move Theorem

We have seen that a stone i is playable if $C(i) = 0$. A move is non-suicidal if $L(i) \neq \varnothing$. To detect a capture, we must tentatively simulate playing at i to verify whether there exists any adjacent opposing stone or group $X \in K(i)$ whose group is thereby left with no liberties. This occurs if $L(X) = \varnothing$; since the sole liberty remaining to that group was precisely the intersection just played, occupying it effects the capture.

Had $L(X) = \varnothing$ held prior to the move, the group would already have been removed from the board.

$$f(n) = i \text{ results in a capture} \iff \exists X \in K(i) : L(X) = \varnothing.$$

The detection of *Superko* requires considerably more extensive analysis, which we address in Section 2.4.

We define the Boolean functions Vacant, Suicide, Capture, and SKo. These functions take a candidate move $f(n) = i$ as their argument and return 1 if the condition is satisfied, and 0 otherwise:

$$\text{Vacant}(f(n)) = \begin{cases} 1 & \text{if } C(i) = 0, \\ 0 & \text{if } C(i) \neq 0; \end{cases} \qquad \text{Suicide}(f(n)) = \begin{cases} 1 & \text{if } L(i) = \varnothing, \\ 0 & \text{if } L(i) \neq \varnothing; \end{cases}$$

$$\text{Capture}(f(n)) = \begin{cases} 1 & \text{if it captures,} \\ 0 & \text{otherwise;} \end{cases} \qquad \text{SKo}(f(n)) = \begin{cases} 1 & \text{if it breaks the rule,} \\ 0 & \text{otherwise.} \end{cases}$$

With these functions, we can formally characterize when a move is legal.

Theorem 2.2.1 (Legal Move Theorem)**.** *Let $f(n) = i$ be a move at turn n of a Go game. Said move is legal if and only if it satisfies the following logical proposition:*

$$f(n) \textit{ is legal} \iff \textit{Vacant} \wedge \neg \textit{SKo} \wedge (\neg \textit{Suicide} \vee \textit{Capture}).$$

Proof. ($\Rightarrow$) Suppose that $f(n)$ is a legal move. By definition, a legal move does not violate the *Superko* rule; hence $\text{SKo}(f(n)) = 0$, and consequently $\neg\text{SKo}(f(n)) = 1$. Furthermore, we know that the *suicide rule* prohibits suicidal moves unless they result in a capture. That is, either the move is not a suicide, or (if it is) it necessarily effects a capture. In logical terms, since $f(n)$ is legal, the proposition $\neg\text{Suicide} \vee \text{Capture}$ is always true.

The move must be placed on a vacant point on the board; that is, $\text{Vacant}(f(n)) = 1$.

There exist only three categories of legal moves:

1. The intersection is vacant, there is no *Superko*, there is no suicide, and no stones are captured.
2. The intersection is vacant, there is no *Superko*, there is no suicide, and a capture occurs.
3. The intersection is vacant, there is no *Superko*, there is suicide, but a capture occurs.

($\Leftarrow$) Let us examine the truth table of our proposition. We denote Suicide by S and Capture by Cap for readability. We omit the cases where *Superko* is violated (SKo = 1) or where the intersection is occupied ($C(i) \neq 0$), as they are trivially false.

Vacant	$\neg$SKo	$\neg S$	Cap	$\neg S \vee$ Cap	Vacant $\wedge \neg$SKo $\wedge (\neg S \vee$ Cap)
1	1	1	1	1	1
1	1	1	0	1	1
1	1	0	1	1	1
1	1	0	0	0	0

The truth table demonstrates that the only rows rendering the full proposition true correspond precisely to the three configurations deduced in the forward direction. ●

Furthermore, we have characterized not only when a move is legal, but also which types of moves exist:

1. Placing a stone without capturing.
2. Executing a capture.
3. Executing a capture by means of suicide.

2.2.4 Scoring

Eventually, by virtue of Superko and mutual agreement, the game concludes. The winner in Go is the player who amasses the greatest number of points. To compute this, it suffices to sum each player's living stones remaining on the board at the end of the game and add their secured territory. In addition, the *komi* is added to White's total.

If we define $P(2)$ as White's score and $P(1)$ as Black's score, we obtain:

$$P(1) = \sum_{I_1 \in I^1} |I_1| + \sum_{J_1 \in J^1} |J_1|, \quad P(2) = \sum_{I_2 \in I^2} |I_2| + \sum_{J_2 \in J^2} |J_2| + \text{komi}.$$

To determine the winner, it suffices to evaluate the difference. If $P(2) - P(1) \geq 0$, White wins; otherwise, Black wins.[10]

[10]We employ $\geq$ to account for the possibility of playing without *komi* or with a *komi* of 0 points, thereby allowing for a draw.

2.3 Go in n Dimensions and with m Players

If we transition to an n-dimensional space, we require n sets to construct the board. As before, for each dimension i, given $n_i \in \mathbb{N}$, we define $N_i = \{1, 2, \ldots, n_i\}$. The board is then formed by the Cartesian product:

$$G = N_1 \times N_2 \times \cdots \times N_n.$$

For m players, we extend the set of states to $E = \{0, 1, \ldots, m\}$, always reserving the value 0 for vacant intersections.

Definition 2.3.1 (Adjacency Between Points)**.** Let T be a board, and let $t_1, t_2 \in T$ be two points given by $t_1 = (a_1, a_2, \ldots, a_n)$ and $t_2 = (b_1, b_2, \ldots, b_n)$.

We say that t_1 is adjacent to t_2, denoted by $t_1 \sim t_2$, if:

$$\exists!\, i \in \{1, 2, \ldots, n\} : \big(\forall r \neq i,\ a_r = b_r \ \wedge\ a_i = b_i \pm 1\big).$$

When reduced to two dimensions, the condition demands one identical coordinate and another differing by exactly 1, naturally recovering the definition of the classical board. Likewise, we extend the notion of connection between stones.

Definition 2.3.2 (Connection)**.** Let T be a board and let $i_1, i_2 \in T$ be two stones such that $i_1 = (e, t_1)$ and $i_2 = (e', t_2)$. These are connected if:

$$i_1 \leftrightarrow i_2 \iff t_1 \sim t_2 \text{ and } C(i_1) = C(i_2).$$

From now on, whenever we refer to this type of connection, we shall call it *orthogonal adjacency*, since we will see that it is possible to employ other criteria in Part III.

The definitions of groups remain identical, as they depend solely on the connection between stones. In turn, territories and captures function in exactly the same manner, since we only require the adjacencies of the group. The only element that truly changes upon introducing more players is the capture criterion; more specifically, the manner in which points are distributed upon removing opposing stones.

Before proceeding, let us highlight a geometric property of vital importance for the subsequent sections.

Lemma 2.3.1. *On an n-dimensional Go board, with the orthogonal adjacency, a point $t = (a_1, a_2, \ldots, a_n)$ has at least n and at most $2n$ adjacencies.*

Proof. For each coordinate of t, there exists at least one valid direction, since if one is blocked by the edge, the opposite direction remains available. Consequently, each dimension contributes at least one adjacency. At an interior point, each coordinate contributes exactly two. Therefore, there are at least n adjacencies and at most $2n$. ●

To simplify notation, we shall adopt the convention of denoting Go in n dimensions and with m players as Go_m^n. Should we wish to explicitly specify the size of each dimension, we may do so in the following manner: $\mathrm{Go}_2^2(19, 19)$ for the case of the standard board. If the number of players is omitted, it is to be understood that we are referring to a generic n-dimensional board, denoted by Go^n. Likewise, Go_m will represent a board of arbitrary[11] dimension played by m players.

[11] The infinite theory of Go is yet to be developed; I am referring to numerable dimension.

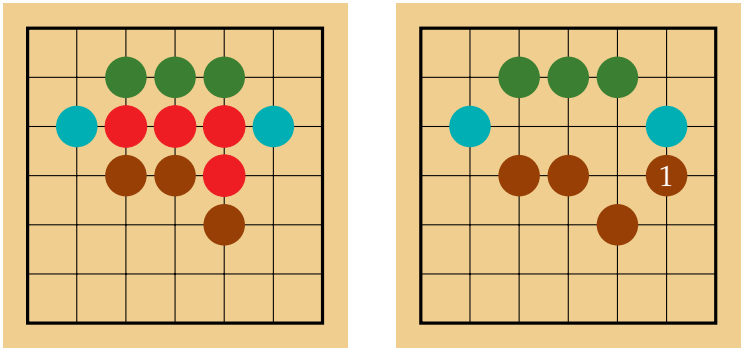


Figure 2.9 – Capture in Go_4^2. Brown plays and captures the red group.

2.3.1 Captures and Points for m Players

Let I be a group of stones. Just as on the classical board, for said group to be considered captured, it must hold that $L(I) = \varnothing$. Consider the Go_4^2 board illustrated in Figure 2.9. Let us suppose our players are: cyan (c), red (r), green (v), and brown (m). Our state set is then defined as $E = \{0, c, r, v, m\}$.

In this example, we observe that a group of 4 red stones is captured after the brown player places their stone at position $(6, 4)$. However, we note that the red group was not completely surrounded exclusively by brown stones. Of the 9 adjacent intersections (the liberties originally possessed by the red group), 3 have been occupied by the green player, 2 by the cyan player, and, finally, 4 by the brown player.

How do we distribute the capture points in this scenario? Our proposal is to distribute the points in strict proportion to the influence each color had over the captured group. If the total to be distributed is the 4 points corresponding to the red stones, we allocate them according to the fraction of liberties occupied by each attacker:

$$\text{Green's points} = \frac{3}{9} \cdot 4, \quad \text{Cyan's points} = \frac{2}{9} \cdot 4, \quad \text{Brown's points} = \frac{4}{9} \cdot 4.$$

Indeed, the sum of the parts equals the total number of points:

$$\left(\frac{3}{9} + \frac{2}{9} + \frac{4}{9}\right) \cdot 4 = 4.$$

In this manner, a proportional distribution is achieved, rewarding the player who contributes most to the capture. Awarding the points solely to the player who executes the capture would disincentivize aggression; it would be unfeasible to surround an opponent if, during the $m - 1$ intermediate turns, others could "steal" the capture.

2.3.2 Komi for Multiple Players

The origin of *komi* lies in preventing draws and counterbalancing the disadvantage of playing second; therefore, it is necessary to reinterpret this concept when multiple players are involved. What follows is a proposal on how we might generalize it.

Stepped Komi

We can define *komi* as a function that depends strictly on the turn order. If we consider the set of players $E \setminus \{0\}$, we can establish a compensation constant $\alpha \in \mathbb{Q}$. Thus, the

komi function $K : E \setminus \{0\} \to \mathbb{R}$ is defined as:

$$K(j) = (j - 1) \cdot \alpha.$$

Irrational Komi

If we additionally seek to preserve the second historical function of *komi*—preventing draws—we may proceed as follows. To guarantee that no two players achieve the same final score, we can assign a unique irrational component to the *komi* of each participant. Given that both the counting of stones and territories (integers) and the proportional captures (rationals) remain strictly within $\mathbb{Q}$, the addition of a specific irrational value for each player eliminates any possibility of a tie. Because the sum of a rational and an irrational is always irrational, equality between two scores is impossible, as their irrational components are always distinct.

Auction Komi

Prior to the start of the game, each player $j \in E \setminus \{0\}$ secretly declares a bid $b_j \in \mathbb{Q}$, representing the number of points they are willing to sacrifice from their final score in exchange for the privilege of securing the initiative (i.e., a negative *komi*). Once all bids have been collected, they are revealed. The turn order is established in strictly decreasing order: the player with the highest b_j value will play first, followed by the second highest, and so forth. In the event of a tie between two or more bids, the turn order for those players is resolved in favor of whoever placed their bid first.

These solutions may be employed simultaneously or further refined; what is presented here are ideas on how to rectify the lack of a well-defined *komi* in generalized Go.[12]

2.4 Positional Superko Detection

By virtue of the Legal Move Theorem 2.2.1, which by construction holds for n dimensions, we possess the theoretical tool to determine whether a move is legal; however, thus far we have not discussed the mechanism for its practical detection.

We have already formalized how to determine whether a capture or a suicide occurs, or whether the intended intersection is occupied. Now, when does a *Superko* occur? That is to say, how do we effectively ascertain that we have violated this rule?

A violation of the *Superko* rule implies that the candidate move recreates a board configuration that has already occurred in the game. To detect this, it would suffice to maintain a historical record of the states of every intersection at each turn. Thus, by tentatively simulating the intended move and comparing the resulting state against the history, we could determine that the rule has been broken if an exact match exists.

To solve this problem, we shall make use of the work developed by Albert L. Zobrist[12].

[12] Author's note: In my opinion, *komi* should not exist; the initiative (*sente*) can be lost at any moment. Furthermore, the larger the board, the less it matters who plays first. This 20th-century invention is only useful for professional players.

2.4.1 Zobrist Hashing

For this method, we shall make use of the logical *exclusive or* operator, denoted as XOR ($\oplus$), applied to random numbers r_i represented as n-bit integers.

An n-bit integer is a sequence of n binary digits. For example, the number 15 in 4 bits is represented as 1111, since:

$$2^3 \cdot 1 + 2^2 \cdot 1 + 2^1 \cdot 1 + 2^0 \cdot 1 = 8 + 4 + 2 + 1 = 15.$$

Likewise, the number 9 in 4 bits is 1001. Let us recall the truth table for the logical XOR operator (see Figure 2.10).

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Figure 2.10 – XOR truth table.

Performing the XOR operation between 15 and 9 in base 2 means applying the operator individually to each corresponding bit. It is worth clarifying that, although we write the numbers contiguously, we actually treat them as if they were, in this case, 4-tuples: $1111 \oplus 1001 = 0110$. If r_i and r_j are two random sequences of n-bit integers, we shall assume that the following properties hold:

P1) $(r_i \oplus r_j) \oplus r_k = r_i \oplus (r_j \oplus r_k)$. (Associativity)

P2) $r_i \oplus r_j = r_j \oplus r_i$. (Commutativity)

P3) $r_i \oplus r_i = 0$. (Nilpotence)

P4) If $s = r_1 \oplus r_2 \oplus \cdots \oplus r_k$, then s is also a random sequence.

P5) The resulting values s are uniformly distributed.

To begin constructing our algorithm, we must take the set of all possible configurations of a stone on the board, namely, $E \setminus \{0\} \times G$. We exclude 0 since we do not place stones of color 0 during play. Next, we assign a unique, random n-bit integer to each element. If we recall that a player has the option to *pass* their turn (a move that does not alter the board state), this action will be assigned $\vec{0}$, which guarantees that the state remains unchanged (as $\vec{0}$ is the neutral of the sum).

To initialize the hash, we must guarantee more numbers than positions. Knowing the cardinality $|E \setminus \{0\} \times G|$ and adding 1 for the empty board, the minimum length of n is:

$$2^n \geq |E \setminus \{0\} \times G| + 1.$$

For our case, on the standard board, we will have 361 intersections and 2 possible states for the stones (white and black). Therefore:

$$|E \setminus \{0\} \times G| = 2 \times 361 = 722.$$

Hence, the minimum number of bits required to map the basic board will be $n = 10$, since

$$2^{10} = 1024 \geq 722 + 1.$$

Zobrist warns of the existence of two types of errors. A *Type 1 error* occurs when two distinct board configurations produce exactly the same hash value; in the context of *Superko*, this would result in a false positive. On the other hand, a *Type 2 error* occurs due to memory space limitations, which we shall not consider. To reduce the probability of a Type 1 error, Zobrist proposes two complementary solutions. The first consists of generating a second hash simultaneously, utilizing an independent sequence, to rule out false positives. The second solution entails using a greater number of bits than strictly necessary. If n is the number of bits we intended to use and m is the number of bits to add, Zobrist demonstrates that the probability of no Type 1 errors occurring is given by the following expression (see Table 2.1):

$$p(\text{no error}) = \left(1 - \frac{1}{2^{n+m}}\right)^{2^n} \approx \left(\frac{1}{e}\right)^{2^{-m}}.$$

m **(extra bits)**	**Prob. of no error**	m **(extra bits)**	**Prob. of no error**
0	0.36778	6	0.98449
1	0.60653	7	0.99222
2	0.77879	8	0.99611
3	0.88251	9	0.99805
4	0.93941	10	0.99901
5	0.96924		

Table 2.1 – Probability of no error by using more bits.

Therefore, it would suffice to utilize 21-bit or even 32-bit numbers to achieve a probability of avoiding a Type 1 error of 0.9999995 or 0.9999999997, respectively. We could generate two, three, or however many verification sequences we deem necessary in order to minimize the error.

To verify the *Superko* rule, it suffices to perform $S \cdot (t-1)$ operations, where S is the total number of independent hash sequences implemented. If $H_s(t)$ represents the hash value at turn t for sequence s, then:

$$\text{SuperKo at turn } t \iff H_s(t) \oplus H_s(t-k) = \vec{0} \text{ and } f(t), f(t-k) \neq \varnothing.$$

for some $k \in \{1, 2, \ldots, t-1\}$ and for every sequence $s \in \{1, \ldots, S\}$. We shall denote the numbers assigned to the stones by $h_s(i)$. Furthermore, to process captures, we shall make use of the XOR operator properties assumed previously. When a stone is captured, it is removed from the board. Therefore, if stone p_j is removed, it suffices to apply the operation $H_s(t) \oplus h_s(i_j)$ to the current state.

By virtue of Properties P1 and P2, we can rearrange the hash calculation:

$$\begin{aligned}
H_s(t) \oplus h_s(i_j) &= \big(h_s(i_1) \oplus \cdots \oplus h_s(i_j) \oplus \cdots \oplus h_s(i_m)\big) \oplus h_s(i_j) \\
H_s(t) \oplus h_s(i_j) &= h_s(i_1) \oplus \cdots \oplus \big(h_s(i_j) \oplus h_s(i_j)\big) \oplus \cdots \oplus h_s(i_m) && \text{(P1 and P2)} \\
H_s(t) \oplus h_s(i_j) &= h_s(i_1) \oplus \cdots \oplus \vec{0} \oplus \cdots \oplus h_s(i_m). && \text{(P3)}
\end{aligned}$$

Given that the null vector $\vec{0}$ is the identity element of the operation ($\vec{0} \oplus h(i_p) = h(i_p)$), the term vanishes. Thus, we have demonstrated that removing a stone (capturing) is equivalent to adding its hash once again. To conclude this chapter, let us illustrate with a concrete example on a small $\text{Go}_2^2(3,3)$ board.

Since the board possesses 9 intersections, we have $|(E \setminus \{0\}) \times G| = 18$. Hence, the minimum number of bits n required will be 5, because $2^5 = 32 \geq 18+1$. If we decide to append 3 additional bits as a safety margin, we will have 8-bit numbers. To initialize, it suffices to randomly assign a number to each possible state at every intersection. To this end, we construct the following table:

Stone	Number	Stone	Number	Stone	Number
(1,(1,1))	01111111	(1,(2,1))	00010110	(1,(3,1))	01001101
(1,(1,2))	10001101	(1,(2,2))	11110001	(1,(3,2))	11101111
(1,(1,3))	10111110	(1,(2,3))	10110001	(1,(3,3))	00011110
(2,(1,1))	00001110	(2,(2,1))	00000010	(2,(3,1))	01001110
(2,(1,2))	10010011	(2,(2,2))	10111010	(2,(3,2))	01000111
(2,(1,3))	10110111	(2,(2,3))	10010001	(2,(3,3))	11001100

Table 2.2 – Hash table of each point in the 3×3 board.

We initialize the game hash to the null vector $\vec{0} = 00000000$, representing the empty board. From this point onward, for every executed move, we update the state by applying the XOR operation ($\oplus$) between the hash of the current turn and the value of the newly placed stone (or removed stone, in the event of a capture). Let us suppose that the following sequence of moves is executed, yielding the board shown in Figure 2.11:

$$f(1) = (1,(2,2)), \quad f(2) = (2,(1,3)), \quad f(3) = (1,(2,3)), \quad f(4) = (2,(2,1)),$$
$$f(5) = (1,(3,1)), \quad f(6) = (2,(3,2)), \quad f(7) = (1,(1,2)), \quad f(8) = \varnothing,$$
$$f(9) = \varnothing.$$

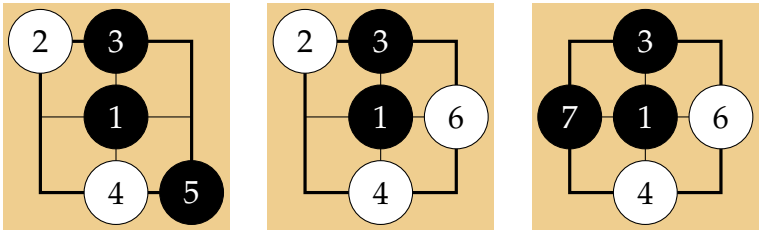


Figure 2.11 – Finished game on a 3×3 board.

We shall compute the states turn by turn, partitioning the vectors into 4-bit blocks to facilitate readability. Denoting the hash of a stone by $h(p_i)$, where $p_i = (e,t)$, the evolution of the game proceeds as follows. On the sixth move, a capture is executed upon the black stone at $(3,1)$; consequently, in addition to applying the hash of the newly played stone, we must apply the hash of the removed stone to "undo" its presence. Should multiple stones be captured, it would suffice to iterate the operation for each of them in any arbitrary order (by virtue of P1 and P2):

$$
\begin{aligned}
H(0) &= \vec{0} \\
H(1) &= H(0) \oplus h((1,(2,2))) = 1111\,0001 \\
H(2) &= H(1) \oplus h((2,(1,3))) = 0100\,1111 \\
H(3) &= H(2) \oplus h((1,(2,3))) = 1011\,1110 \\
H(4) &= H(3) \oplus h((2,(2,1))) = 1010\,1000 \\
H(5) &= H(4) \oplus h((1,(3,1))) = 0110\,0101 \\
H(6) &= H(5) \oplus h((2,(3,2))) \oplus h((1,(3,1))) = 1100\,0111 \\
H(7) &= H(6) \oplus h((1,(1,2))) \oplus h((2,(1,3))) = 1111\,0100 \\
H(8) &= H(7) \oplus 0000\,0000 = H(7) \\
H(9) &= H(8) \oplus 0000\,0000 = H(8).
\end{aligned}
$$

Figure 2.12 – Game hashes; moves in blue, captures in red.

At turns 8 and 9, passing the turn produces a state identical to the preceding one. This does not violate the *Superko* rule, as the rule solely prohibits placing a stone to recreate a position.

However, it could occur that $h(2,(3,2)) = 1100\ 1011$. This would trigger a false positive for *Superko* on the sixth turn, since $H(6) = H(5) \oplus h(2,(3,2)) = 1011\ 1110$ and, upon comparing it with the history,

$$H(6) \oplus H(3) = 0000\ 0000.$$

Thus, by employing additional bits and implementing multiple simultaneous sequences to rule out collisions, Zobrist's method is fully implemented in Go.

2.5 The Structure of Time

We are now in a position to inject time into the structure. We define $\mathcal{T}$ as the space of all boards,

$$\mathcal{T} = \{T \subset E \times G \mid T \text{ is a board}\}.$$

2.5.1 The Game as a Trajectory

A game constitutes a sequence of moves $f : \mathbb{N} \to \mathcal{J}$, where $f(t)$ represents the action chosen by the active player at turn t. This function f naturally induces a sequence of boards in $\mathcal{T}$, which we shall term a trajectory:

$$\{T_0, T_1, \ldots, T_{end}\}$$

The sequence is constructed recursively by means of Δ:

1. **Initial state ($t = 0$):** T_0 is the empty board; that is, the set $T_0 = \{(0,t) : t \in G\}$.
2. **Transition ($t \to t+1$):** The board at the subsequent turn is the result of applying the transition to the move of the current turn:

$$T_t = \Delta(T_{t-1}, f(t))$$

2.5.2 State Dynamics

Under this construction, we formalize the properties of the game's flow. We shall denote by $E^k(T_t) = \{i \in T_t : C(i) = k\}$ the set of all stones of color k present on the board at turn t, and by $|E^k(T_t)|$ its cardinality.

- **Growth and Decrease:** If at turn t the active player is $k \in E \setminus \{0\}$, the new board given by $T_t = \Delta(T_{t-1}, f(t))$ will possess, at most, one additional stone of their color. In contrast, the stones of the other colors ($j \neq k$) can only decrease or remain constant due to potential captures:

$$|E^k(T_t)| \leq |E^k(T_{t-1})| + 1,$$
$$|E^j(T_t)| \leq |E^j(T_{t-1})|, \quad \forall j \neq k.$$

- **Halting Condition:** Let $m = |E \setminus \{0\}|$ be the total number of players. The game concludes at turn T_{end} when all players pass consecutively. This occurs when:

$$f(T_{end}) = f(T_{end-1}) = \cdots = f(T_{end-m+1}) = \varnothing$$

  Consequently, the state of the board freezes, resulting in:

$$T_{end} = T_{end-1} = \cdots = T_{end-m}$$

  This state T_{end} constitutes the Final Board.

2.5.3 Anatomy of a Move

Let us understand how Δ functions. Suppose we are playing Go and that, on our turn, we intend to play $f(t)$. If $f(t) = \varnothing$, the board remains identical and we pass the turn; that is,

$$T_t = \Delta(T_{t-1}, \varnothing) = T_{t-1}.$$

If $f(t) = i^* = (e, p)$, we must validate said move. The computation of the new board $T_t = \Delta(T_{t-1}, i^*)$ by virtue of Theorem 2.2.1 is performed as follows:

1. **Placement:** If there exists $(e', p) \in T_{t-1}$ such that $e' \neq 0$, the move is illegal. If $e' = 0$, we generate a temporary board T' where intersection p, previously empty, acquires state e. We then proceed to the next step.

$$T' = \big(T_{t-1} \setminus \{(0, p)\}\big) \cup \{(e, p)\}.$$

2. **Capture Detection:** Evaluating the adjacencies in T', we identify the set Cap by taking the union of all opposing groups that possess no liberties.

$$Cap = \bigcup \{i \in I^{e'} \subset T' \mid L(I^{e'}) = \varnothing, e' \notin \{0, e\}\}.$$

   If $Cap \neq \varnothing$, the move is legal and we skip to step 4; otherwise, we proceed to the next step.

3. **Suicide Detection:** Considering $Cap = \varnothing$, we evaluate the state of the played stone $i^* = (e, p)$ in T'. The analysis is divided into two cases:

   - **Isolated stone:** If i^* does not connect to any stone of its own color, we evaluate its liberties. If $L(i^*) = \varnothing$, the move constitutes a suicide.
   - **Union with a group:** Let I be the group to which i^* belongs on this new board. If $L(I) = \varnothing$, the move constitutes a suicide.

   If either of these two scenarios occurs, the move is illegal. If there is no suicide, we proceed to the next step.

4. **Update:** We transform the captured stones to the free state (Cap_0) and update the board.

$$Cap_0 = \{(0, P(i)) : i \in Cap\}, \quad T_t = (T' \setminus Cap) \cup Cap_0.$$

   We then proceed to the next step.

5. **Superko Detection:** We compute $H(T_t)$ and compare it against the history. If the comparison yields $\vec{0}$ for any hash, we have violated the rule and the move is discarded (in every hash sequence); otherwise, the turn concludes.

In this manner, any candidate move can be accepted or rejected. A Game of Go is formally defined as a 4-tuple,

$$\Gamma = (G, E, A, \Delta).$$

where, to a Go game (G, E, A), we adjoin the transition function $\Delta : \mathcal{T} \times \mathcal{J} \to \mathcal{T}$.

2.6 Life in the Game of Go

If we recall the rules, towards the very end we briefly encountered the concept of being *alive*. In this section, we shall formalize this notion.

In Figure 2.13, White cannot capture because the suicide rule prohibits them from playing on Black's remaining liberties. The algorithm for determining unconditional life is straightforward: we simulate Black passing while White attempts to occupy their liberties. If White fails to capture it, the group possesses unconditional life.

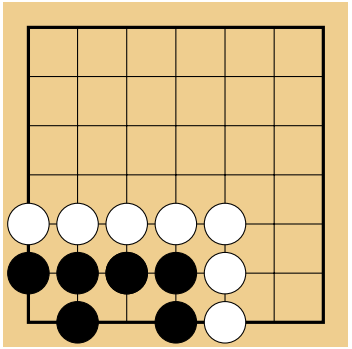


Figure 2.13 – Board with a black group that has unconditional life.

In 1976, David B. Benson formalized this concept [1]. He demonstrated that the survival of a set of stones depends solely on its adjacencies. In what follows, we shall adapt his definitions and theorems to our formal framework.

To achieve this, we first need to count how many groups surround a region. Let P be a set of stones of color $e \in E \setminus \{0\}$, and let R be a region completely surrounded by P (that is, $K(R) \subseteq P$). We are interested in determining, knowing only R, how many distinct groups comprise its *boundary* in P.

We know that the set $K(R)$ contains the stones adjacent to the region. As seen in Part I, $0 \notin C(K(R))$.

Definition 2.6.1 (Boundary of a region)**.** Let R be a region. We define the boundary[13] of R, denoted by ∂R, as the set of all groups or stones that have a non-empty intersection with the adjacencies of the region:

$$\partial R = \{I \mid I \cap K(R) \neq \varnothing\}$$

It is crucial to note the difference: ∂R collects the adjacent groups, whereas $K(R)$ collects the adjacent stones; Figure 2.14 illustrates this distinction.

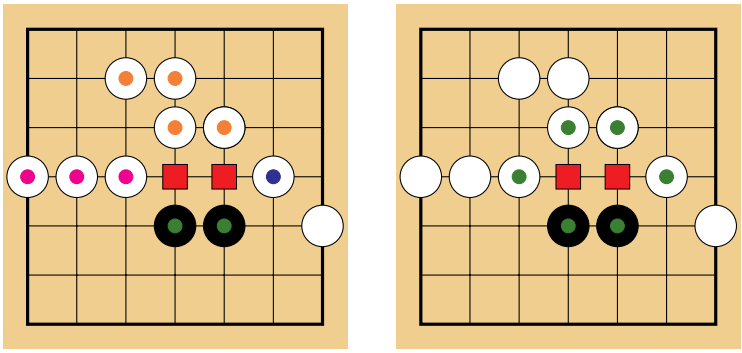


Figure 2.14 – Left: Board with the *boundary groups* of a Region; Orange, Magenta, Green, Blue. Right: Adjacencies of a Region.

The cardinality of this set indicates exactly how many independent groups and stones surround the empty region.

However, quantity alone does not suffice; we must also evaluate the quality of these regions. In Benson's terminology, they must be *healthy* for the groups that contain them.

Definition 2.6.2 (Healthy region)**.** Let P be a set of groups and stones of the same color, and let R be a region surrounded by P (that is, $\partial R \subset P$). We say that R is healthy for a group $I \in P$ (or $i \in P$) if every element of R is a liberty for I (i). Formally:

$$R \text{ is healthy for } I \in P \iff R \subseteq L(I).$$

This condition ensures that if the opponent attempts to play within the region, they will immediately come into contact with the group I (i). In the example of Figure 2.15, the intersections marked in green do not satisfy this premise; therefore, that region is not healthy for any black group.

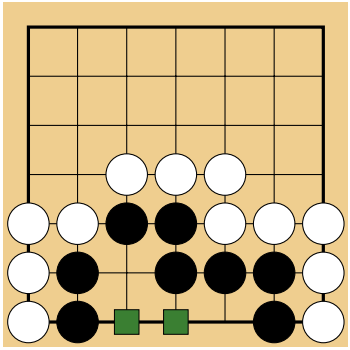


Figure 2.15 – Board with black groups whose controlled region is unhealthy.

Definition 2.6.3 (Vital region)**.** Let P be a set of groups and stones of the same color, let $I \in P$ be a specific group, and let R be a region. We shall say that R is vital for I in P if the following conditions are simultaneously satisfied:

[13] Benson calls this the Neighborhood

1. R is healthy for I; $R \subseteq L(I)$.

2. All groups comprising the boundary of R belong to P; $\partial R \subseteq P$.

We shall denote vital regions by V. This definition formalizes the requirement that a region must provide immediate liberties (be healthy) and be surrounded by groups of the same color. Building upon the notion of a vital region, we can classify the nature of completely surrounded regions.

Definition 2.6.4 (Eye)**.** Let V be a vital region for a group I (or stone) within a set P. If the boundary of V is comprised of a single independent group—that is, if $|\partial V| = 1$—then V constitutes an eye for I.

When the region is surrounded by multiple distinct groups of the same color, it constitutes a *union*.

Definition 2.6.5 (Union)**.** Let R be a healthy region for a group I (or stone) within a set P. If the boundary of R is composed of more than one independent group—that is, if $|\partial R| > 1$—then R constitutes a union for the set P (a "*joiner*" in Benson's terminology). Figure 2.16 illustrates the subtle difference between Benson's definitions.

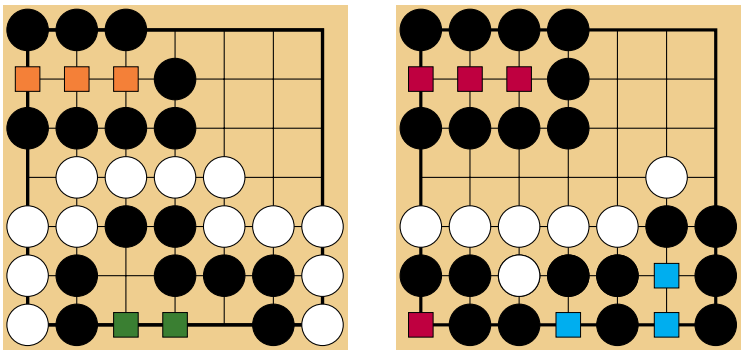


Figure 2.16 – Left: healthy and unhealthy. Right: eyes and unions.

2.6.1 Benson's Theorem

Let us apply the theory and examine Benson's Theorem. Before proceeding to the main theorem, we must establish four preliminary lemmas. In what follows, we shall assume that the player defending the attacked group passes on every turn.

Lemma 2.6.1. *If a group I possesses at least two distinct eyes, then it possesses unconditional life.*

Proof. Let O_1 and O_2 be the eyes of I. To capture group I, the opponent must occupy all of its liberties, which entails completely filling O_1 and O_2.

Without loss of generality, let us suppose that the opponent attempts to fill O_1 first. Since the region is finite, there will eventually remain a single intersection $x \in O_1$.

Upon playing at x, the opponent's stones within O_1 are left with no liberties, as O_1 is exclusively surrounded by I ($|\partial O_1| = 1$). Given that I still possesses at least one

liberty in O_2, the move at x does not capture I. Consequently, the suicide rule dictates that occupying intersection x is illegal.

As a result, it is impossible to completely eliminate either of the two eyes by means of legal moves. Therefore, I possesses unconditional life. ●

Lemma 2.6.2. *Let S be a set of groups of the same color such that $|S| \geq 2$. If a group $I_1 \in S$ possesses an eye O and at least one union U whose boundary contains another distinct group $I_2 \in S$, then I_2 must be captured before I_1 can be captured.*

Proof. To capture I_1, the opponent must occupy all the liberties provided by the union U, since those of the eye O can only be filled last (Lemma 2.6.1).

Given that U is a union, its boundary contains more than one group ($|\partial U| > 1$), including I_1 and I_2. Every move within U simultaneously reduces the liberties of both. Upon attempting to occupy the last empty liberty of U, the opponent's move will lack liberties of its own. By the suicide rule, this move is legal only if it executes a capture.

Since I_1 retains at least one liberty in its eye O, it cannot be captured on this turn. Therefore, the move must necessarily capture another group on the boundary of U, in this case I_2. Consequently, I_2 is captured before I_1. ●

In summary, to capture a group that possesses an eye and is connected to a union, it is necessary to first capture the other groups comprising the boundary of said union, as illustrated in Figure 2.17.

Lemma 2.6.3. *Let I be a group possessing no eyes, but connected to n unions $U_1, U_2, \ldots, U_n$ ($n \geq 2$). Then, before I can be captured, the opponent must capture at least one distinct group in $n-1$ of these unions.*

Proof. To capture I, the opponent must eliminate all of its liberties across the n unions. Since they cannot simultaneously occupy the final liberty of every union, the suicide rule compels them to reserve at least one union, say, U_n, for the final move.

In order to exhaust the liberties in the $n-1$ remaining unions ($U_1, \ldots, U_{n-1}$), each union presents the situation described in Lemma 2.6.2. Therefore, the opponent will be forced to capture at least one distinct group in each of those $n-1$ unions before being able to attack I. ●

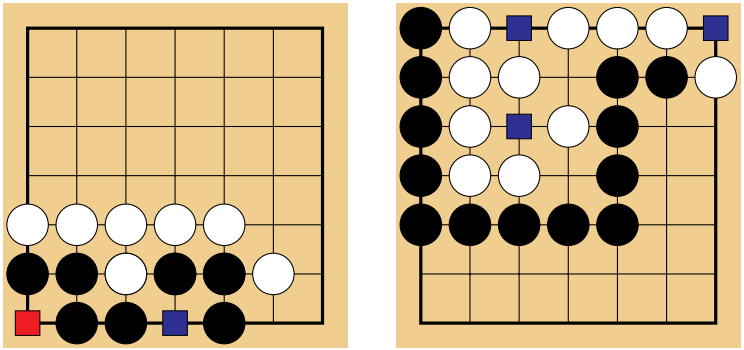


Figure 2.17 – Left: The black group possesses one eye (red) and one union (blue); White must first capture the group on the right (Lemma 2.6.2). Right: The upper-left white group lacks eyes, but shares three unions (blue); Black must capture the neighboring groups in at least two of them before being able to capture it (Lemma 2.6.3).

Lemma 2.6.4. *Let I be a group or stone; the liberties that do not belong to vital regions can be occupied one by one by means of a sequence of legal moves.*

Proof. Let $L(I)$ be the set of liberties of I. Every liberty $x \in L(I)$ belongs to some empty region R. If R is not a vital region for the set P (to which I belongs), this implies that it fails to satisfy the second condition of Definition 2.6.3—that is, its boundary is not entirely contained within P ($\partial R \not\subset P$).

Since there exist elements on the boundary of R that do not belong to P, the opponent can place stones in R without these stones being left with no liberties. Given that $\partial R \not\subset P$, the enemy stones played in the region will retain or find liberties through those groups external to P, thereby preventing the move from constituting a suicide. Therefore, the opponent can occupy these liberties one by one in a legal manner. ●

Theorem 2.6.5 (Benson's Unconditional Life Theorem)**.** *Let I be a group. Under a canonical Go_m game ($m \geq 2$), I possesses unconditional life if and only if it belongs to a set of groups S of the same color such that, for every group $J \in S$, there exist at least two distinct vital regions in S.*

We shall begin by proving the backward implication. Then, via the contrapositive, the forward implication.

Proof. ($\Leftarrow$) : Let $I \in S$. To capture it, we must surround it completely. In particular, we must occupy the liberties of its vital regions. By Lemmas 2.6.1, 2.6.2, and 2.6.3, whether these regions are eyes or unions, the capture of I demands the prior capture of some other group in S that belongs to the boundary of said regions.

Let us denote this group by $I_k \in S$. To capture it, we must eventually occupy the liberties of its own vital regions (which might coincide with those of I). To do so, once again, we must capture another group belonging to the boundary of its vital regions.

Repeating this argument, we deduce that every capture of a group in S requires a strictly prior capture of another group in S. Consequently, no finite sequence of moves can contain a first capture of a group in S. Therefore, S possesses unconditional life and, consequently, so does I.

($\Rightarrow$) : By contraposition, we shall prove that:

$\exists I \in S$ with at most one vital region in $S \implies S$ does not possess unconditional life.

Let $I \in S$ be a group possessing at most a single vital region V in S; by Lemma 2.6.4, it is possible to occupy all the external liberties of I by means of a sequence of legal moves. Once the external liberties are occupied, we analyze the state of I:

- If I lacked vital regions, the occupation of the exterior exhausts its liberties, thereby executing its capture.
- If I possessed exactly one vital region V, all of its liberties lie within V. Upon attempting to occupy the final intersection of V, since I does not possess a second vital region to guarantee it at least one liberty external to the move, placing this stone eliminates the final liberty of I, executing the capture.

In both cases, there exists a valid sequence to reduce the liberties of I and capture it. Therefore, S does not possess unconditional life, concluding the proof. ●

It is crucial to distinguish between being Alive and being Unconditionally Alive. In Figure 2.18, if Black were to play at the green intersection, they would achieve unconditional life; in its current state, the group lacks it. In a game, a group lacking unconditional life may be practically alive, requiring merely one or two moves to secure it. The distinction lies precisely therein: unconditional life implies that, statically, a group cannot die; life in Go, however, is a question of how quickly one can achieve it and whether the other players can prevent it.

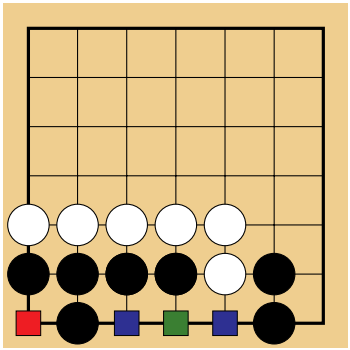


Figure 2.18 – Board with an eye and a union.

2.6.2 Seki

Having examined unconditional life, it is time to address conditional life. There exists another common structure in Go termed *seki*; this occurs when two groups of distinct colors share liberties in such a manner that, if either of them attempts to capture the other, the initiator ends up losing (see Figure 2.19). In Figure 2.19, whoever plays first and takes one of the two liberties of the interior groups loses, since it will no longer be their turn and the other player can (and most likely will) execute the capture.

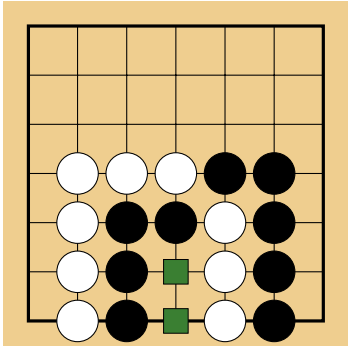


Figure 2.19 – Board with groups in *seki*.

Definition 2.6.6 (Seki without eyes)**.** Let I, J be two groups of distinct colors. If there exists a region R such that $|R| \geq 2$ and $L(I) = L(J) = R$, then I and J are in *seki*.

This definition perfectly encapsulates *seki* without eyes, for if any player attempts to play in R, there will eventually remain a single liberty; whoever plays on the subsequent turn will execute the capture. It is necessary to require that $|R| \geq 2$; were

this not the case, it would mean that both groups possess only a single liberty, and it would suffice for the active player to play on that liberty to execute a capture. By Benson's theorem, a group with two eyes possesses unconditional life. It remains, then, to analyze the case of groups possessing a single eye that find themselves in *seki*.

Definition 2.6.7 (Seki with eyes)**.** Let I and J be two groups of stones of distinct colors. Let O_I and O_J be an eye for I and for J, respectively. If there exists a region $R \subseteq L(I) \cap L(J)$ such that $L(I) = R \cup O_I$ and $L(J) = R \cup O_J$, then I and J are in *seki*.

By Lemma 2.6.1, every eye can be reduced to a single liberty. Consequently, occupying the shared region R entails relying exclusively on said eye, thereby allowing the opponent to capture the group. Although there exist more complex variants of *seki* (false eyes, double *ko*, etc.), these basic configurations suffice to illustrate conditional life.

This section will be continued some time in the future.

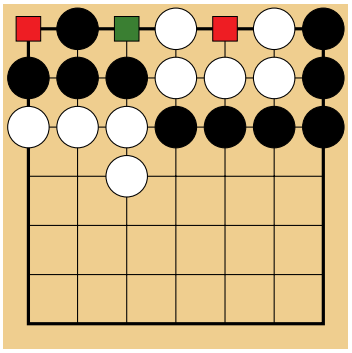


Figure 2.20 – Board with groups in *seki*.

2.6.3 Remark on Kim's Theorem 2

We shall present a counterexample to Kim's Theorem 2 [4, 5]. It is worth noting that Kim first raises the question of embedding an unconditionally alive group from a two-dimensional board into a three-dimensional board. However, Theorem 2 asserts a stronger result,

> "In a 2-person go-game on a 3-dimensional board, every two dimensional x-graph is not safe."

Given that Kim grounds his analysis in Benson's work, within our theoretical framework, this result translates as follows:

"In a Go_2^3 game, no two-dimensional group of color x possesses unconditional life."

For Kim, a "two dimensional x-graph" is a group of color x that holds one of the spatial dimensions fixed at a constant value. It is worth noting that, in his formulation, he imposes a minimum board size of $9 \times 9 \times 9$. We shall exhibit a configuration that contradicts the statement of Kim's Theorem 2.

Due to the three-dimensional nature of the board, we can partition it along its third coordinate into distinct "floors," treating them as $9 \times 9 \times 1$ boards. Let us consider a layer entirely occupied by a black group, which we shall call N, and an entirely empty layer, which we shall call V. Constructing the board from bottom to top along the third coordinate, we can alternate these layers in the following manner, as depicted in Figure 2.21.

$$V, N, V, N, V, N, V, N, V$$

Figure 2.21 – Layers of the counterexample.

Let S be the set formed by all the black groups in the construction. Every intersection of an empty layer is adjacent to a stone from one of the neighboring black layers;

therefore, the region is healthy. Furthermore, every boundary of an empty layer is comprised exclusively of stones belonging to S.

Consequently, they constitute at least two distinct vital regions for each group in S. By Benson's Theorem (Kim's Theorem 1), the set S possesses unconditional life. As a result, Benson's Theorem guarantees the existence of safe groups contained within two-dimensional sub-boards, thereby contradicting Kim's Theorem 2. The construction generalizes immediately to boards of any height. If the number of layers is odd, it suffices to proceed as shown. If it is even, it is sufficient to replace the first black layer with a double layer, consisting of two consecutive layers of black stones, always spacing them with empty layers.

Under the definitions of a vital region and unconditional life adopted by Benson and utilized by Kim, there exist safe two-dimensional groups on a three-dimensional board. This contradicts the statement of Theorem 2.

Furthermore, if we omit the arbitrary board-size restriction imposed by Kim, the phenomenon can be illustrated in its minimal expression with the aid of the 3dGo tool.

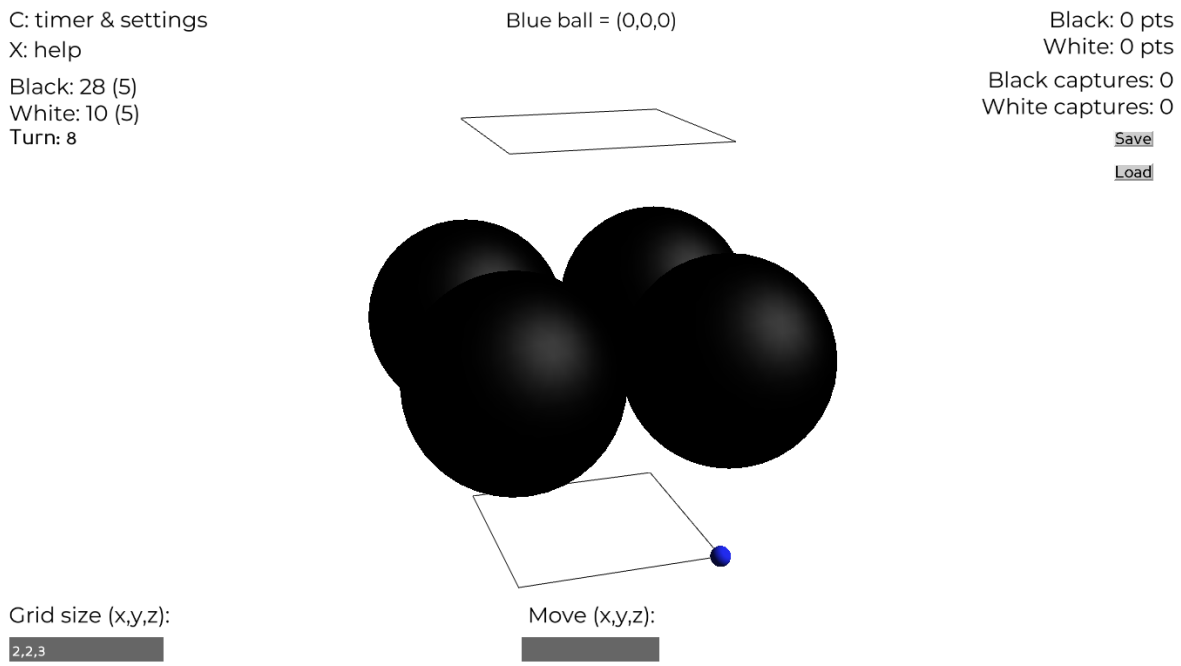


Figure 2.22 – Minimal counterexample to Kim's result. A two-dimensional black group possessing unconditional life on a $2 \times 2 \times 3$ board.

III: Abstraction

井蛙不可以語於海者，
拘於虛也；

夏蟲不可以語於冰者，
篤於時也；

曲士不可以語於道者，
束於教也。

A frog in a well cannot be spoken to about the ocean
because it is confined by its own space.

A summer insect cannot be spoken to about ice
because it is bound by its own season.

A narrow-minded person cannot be spoken to about the Way
because they are bound by their own teachings.

秋水—Zhuāngzǐ, Chapter 17

3.1 Comparing Adjacencies

Observing the two-dimensional board in Figure 3.1, we can note that it will have xy intersections; however, the following question arises: how many adjacencies do all these intersections possess?

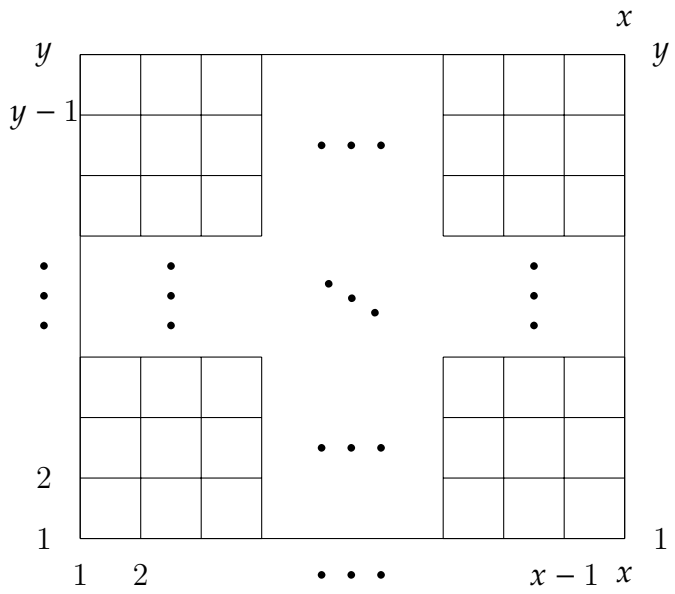


Figure 3.1 – Go board of size $x \times y$.

By Lemma 2.3.1, these nodes have 2, 3, or 4 adjacencies. The 4 vertices possess 2 adjacencies. There are four edges—two of length $(x-2)$ and two of length $(y-2)$—whose nodes have 3 adjacencies. Furthermore, there is an internal region of $(x-2)(y-2)$ nodes, each with 4 adjacencies. This allows us to define the total number of adjacencies on a Go^2 board.

Let $A(x, y)$ be the total number of adjacencies on a board of dimensions $x \times y$.

$$A(x, y) = 2 \cdot 4 + 3(2(x-2)) + 3(2(y-2)) + 4(x-2)(y-2)$$
$$A(x, y) = 8 + 6x - 12 + 6y - 12 + 4xy - 8x - 8y + 16$$
$$A(x, y) = 4xy - 2x - 2y.$$

Let us also define the average number of adjacencies on a board, which follows immediately from the definition of total adjacencies; it suffices to divide by the total number of nodes.

Let $\tilde{A}(x, y)$ be the average number of liberties on a two-dimensional board, then:

$$\tilde{A}(x, y) = \frac{4xy - 2x - 2y}{xy}.$$

3.1.1 Go in Three Dimensions

Let us consider the case for Go^3, consisting of xyz nodes. To this end, we rely on the following diagram, Figure 3.2 (left):

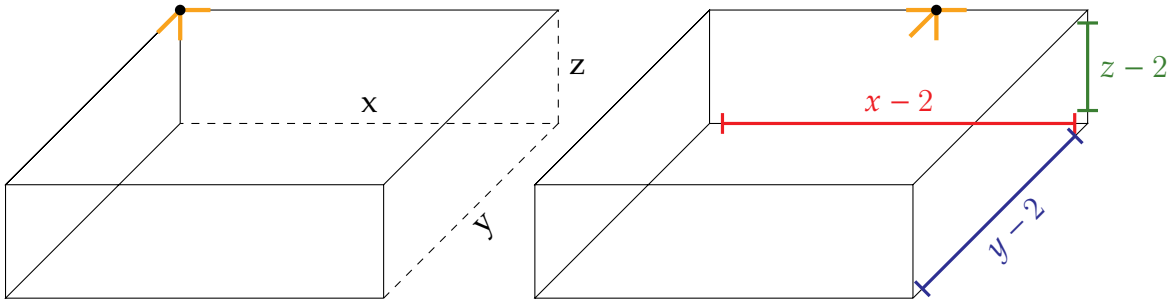


Figure 3.2 – Exterior of the three-dimensional board. A vertex with 3 adjacencies, edges, and a node with 4 adjacencies.

Proceeding as before, we observe that it possesses 8 vertices, each with 3 adjacencies. We identify 4 edges of $(x-2)$ nodes, 4 of $(y-2)$, and 4 of $(z-2)$, each with nodes possessing 4 adjacencies, see Figure 3.2 (right).

Next, we have the faces. These are divided into three pairs: 2 faces of $(x-2)(y-2)$, 2 of $(x-2)(z-2)$, and 2 of $(y-2)(z-2)$, with nodes possessing 5 adjacencies each, Figure 3.3.

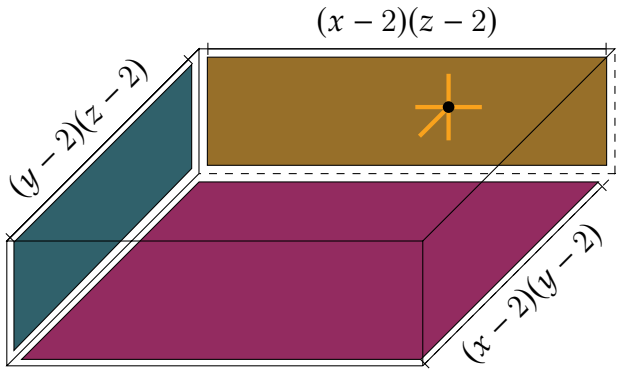


Figure 3.3 – Faces of the three-dimensional board with a node of 5 adjacencies.

Remaining is the internal volume, which consists of $(x-2)(y-2)(z-2)$ nodes, each with 6 adjacencies. We are thus in a position to define the total number of liberties on a three-dimensional board.

Let $A(x, y, z)$ be the number of adjacencies of a three-dimensional board; if:

- $V = 8 \cdot 3$, adjacencies of the vertices.
- $R = 4(4(x-2) + 4(y-2) + 4(z-2))$, adjacencies of the edges.
- $C = 5(2(x-2)(y-2) + 2(x-2)(z-2) + 2(y-2)(z-2))$, adjacencies of the faces.
- $I = 6(x-2)(y-2)(z-2)$, adjacencies in the interior.

In this manner, we can calculate the adjacencies of any three-dimensional board as follows:

$$A(x, y, z) = V + R + C + I$$
$$A(x, y, z) = 6xyz - 2xy - 2xz - 2yz.$$

And its average,

$$\tilde{A}(x, y, z) = \frac{6xyz - 2xy - 2xz - 2yz}{xyz}.$$

In turn, we can define the *exterior* of the three-dimensional board; let $A(x, y, z)_\partial$ be the number of adjacencies of a board of dimensions x, y, z that only takes the exterior into account. As seen previously, it suffices to sum the vertices, the edges, and, in the case of the faces, we no longer count the adjacency pointing toward the interior, since it ceases to exist as there are no interior nodes, such that our faces have 4 adjacencies. It suffices to take $C' = 4(2(x-2)(y-2) + 2(x-2)(z-2) + 2(y-2)(z-2))$, hence:

$$A(x, y, z)_\partial = V + R + C'$$
$$A(x, y, z)_\partial = 8(xy + xz - 2x + yz - 2y - 2z + 3).$$

We immediately define the average by subtracting the nodes we are not considering; let $\tilde{A}(x, y, z)_\partial$ be the average number of liberties on the exterior of a board of dimensions x, y, z:

$$\tilde{A}(x, y, z)_\partial = \frac{8(xy + xz - 2x + yz - 2y - 2z + 3)}{(xyz) - (x-2)(y-2)(z-2)}.$$

3.1.2 Comparing Average Adjacencies

Prior to the development of this work, as a Go player, I wondered whether a three-dimensional version of the game existed. Upon finding only conceptual images on the Internet, I asked myself: why not develop it? Thus, 3dGo was born.

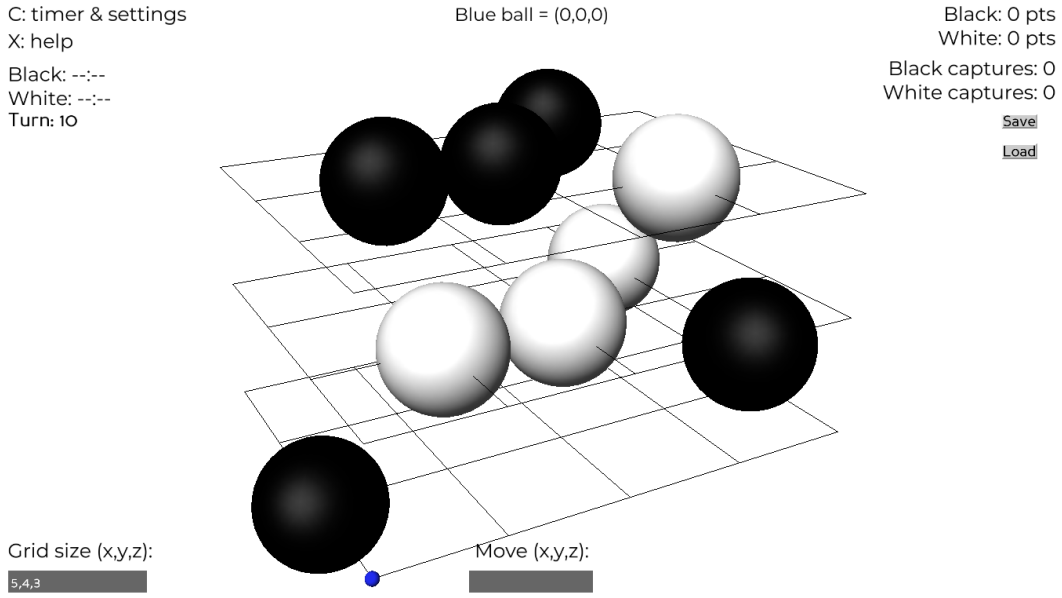


Figure 3.4 – Go board of size $5 \times 4 \times 3$ in 3dGo.

Once the rules, arbitrary board-size selection, and other options were implemented, my opponent and I noticed that it was not the same game. *3dGo* felt freer, more aggressive. The greater number of adjacencies altered the playing style, differentiating it from conventional Go. To understand this difference, even without prior experience, we shall analyze the behavior of the average number of adjacencies in the limit[1]: Let us examine the limits of the average adjacencies for two- and three-dimensional boards.

$$\lim_{x,y\to\infty} \tilde{A}(x,y) = \lim_{x,y\to\infty} \frac{4xy - 2x - 2y}{xy} = \lim_{x,y\to\infty} \left(4 - \frac{2x}{xy} - \frac{2y}{xy}\right) = 4,$$

$$\lim_{x,y,z\to\infty} \tilde{A}(x,y,z) = \lim_{x,y,z\to\infty} \frac{6xyz - 2xy - 2xz - 2yz}{xyz} = 6.$$

It is evident that the average number of adjacencies in Go^3 is considerably higher. This difference is perceptible in the gameplay experience. The following problem then arises: How can we bring this average closer to 4? Playing Go^3 must be equivalent to playing Go; to achieve this, it must faithfully imitate it in rules, size, and gameplay feel. This is why this reduction is necessary, because playing Go^3 is not playing Go^2.

3.1.3 Removing Nodes, Yes. But How?

As we have seen, the majority of the nodes on a Go^3 board lie in the interior—provided the size is sufficiently large—where each possesses 6 adjacencies. Therefore, the elimination of nodes must begin in this region. Next in priority are the faces (5 adjacencies), whereas the edges and vertices, possessing 4 or fewer, do not affect the

[1]This was the motivation for initiating this writing.

average we seek. If we consider only the exterior of the board, the faces possess precisely 4 adjacencies (we lose the one pointing inward); consequently, analyzing their limit, we find that:

$$\lim_{x,y,z\to\infty} \tilde{A}(x,y,z)_\partial = \lim_{x,y,z\to\infty} \frac{8(xy+xz-2x+yz-2y-2z+3)}{(xyz)-(x-2)(y-2)(z-2)} = 4.$$

Thus, our exterior board possesses the same average adjacencies in the limit as Go^2—that is:

$$\lim_{x,y\to\infty} \tilde{A}(x,y) = \lim_{x,y,z\to\infty} \tilde{A}(x,y,z)_\partial.$$

3.1.4 Toroidal Board

In Go^2, the limit of the average as the board is made arbitrarily large is 4. One might ask: is it possible to establish a board whose average liberties are precisely 4?

If we join the opposite edges of the board, we obtain the following, as shown in Figure 3.5.

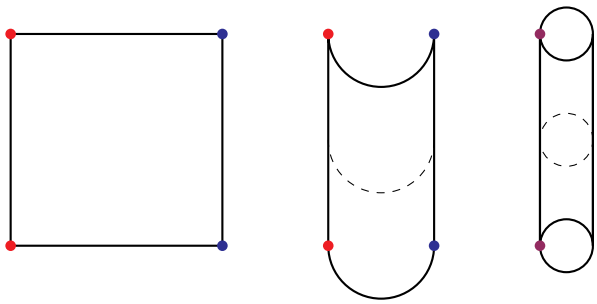


Figure 3.5 – Transition from the plane to a cylinder by joining its opposite sides.

Let us repeat the process; let us join our 4 vertices—which are now 2—into a single one, Figure 3.6.

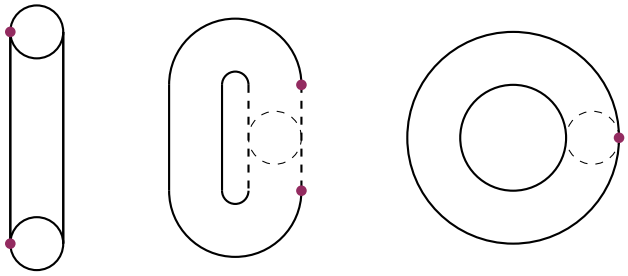


Figure 3.6 – Transition from the cylinder to the torus after joining its opposite faces.

Let us observe how the adjacencies of the vertices have changed: in the plane, we saw that they were 2 for each, but in the cylinder, it can be seen that they have 3, and furthermore, on the torus, they are 4. The same occurs with the nodes on the edges, Figure 3.7.

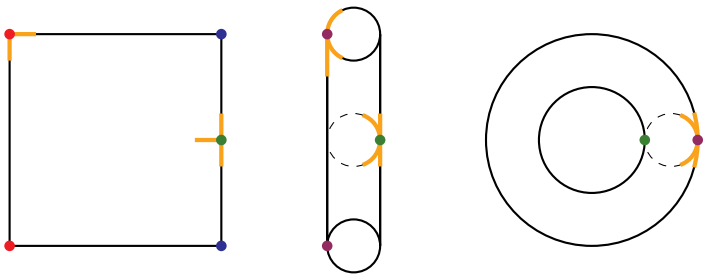


Figure 3.7 – Change in adjacencies from the plane to the torus.

It is therefore worth asking: Is it possible to do this in any dimension? That is, taking them to their "toroidal" form. Indeed, it is possible, and furthermore, it is very simple and, unlike Figure 3.7, we do not lose points.

Under our scrutiny, G contains only points, and the adjacency criterion determines their relationships. To relate the edges, it suffices to add new relationships. What we seek is to join each face with its opposite; to achieve this, we formulate the following definition of the adjacency criterion by opposites, denoted as A_{op}. Let $G = N_1 \times \cdots \times N_m$ be a board, where $|N_k| = n_k$. We shall say that two points $g_1, g_2 \in G$ are oppositely related if they are identical in all their coordinates except in one, the k-th, where they take values at the extremes:

$$g_1 = (x_1, \ldots, 1, \ldots, x_d) \sim (x_1, \ldots, n_k, \ldots, x_d) = g_2.$$

Thus, in this simple manner, we have defined the toroidal board, which is nothing more than the same board as before, except that we are now considering more adjacent points. Formally, if we call the canonical criterion A_{ort} (orthogonal adjacencies), we transition from the form (G, A_{ort}) to $(G, A_{ort} \cup A_{op})$.

It is clear that the toroidal board will have 4 adjacencies for each of its nodes. Let $A(x, y)_\tau$ be the number of adjacencies on a toroidal board of size xy; let us carry out once again the process we performed at the beginning. The toroidal board will have no vertices, no edges, and consequently, a single face composed of xy nodes which, as we have already seen, possess 4 adjacencies per node; thus:

$$A(x, y)_T = 4xy.$$

And its average will be 4, since:

$$\tilde{A}(x, y)_T = 4xy/xy = 4.$$

In this way, we have found a board such that for every node, the number of adjacencies is uniform. Let us do the same in Go^3; we can formulate the following scheme from Figure 3.8.

Here, if we apply the new relation, we obtain that all points will have 6 adjacencies. Indeed, the vertices will acquire 3 new adjacencies, bringing them to 6; the edges will acquire 2 new adjacencies, bringing them to 6; and finally, the faces will acquire a single new adjacency—the opposite one. Thus, equality is achieved in the limit.

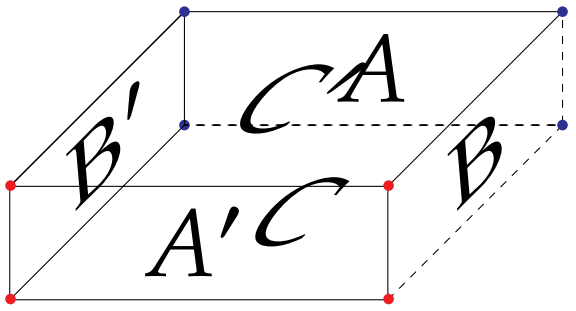


Figure 3.8 – Faces of the Go^3.

Now then, what would happen if we consider only the points lying on the boundary?

If $x \in G$, $x = (x_1, \ldots, x_n)$ with $x_i \leq k_i = |n_i|$, the boundary will be that of the points given by

$$G' = \{x \in G : \exists i, (x_i = 1 \vee x_i = n_i)\}.$$

Through the canonical adjacency criterion, we obtain a board such that it possesses, in the limit, 4 adjacencies per point, since only the vertices possess 3 adjacencies, as we saw in Section 3.1.3.

If we attach to this board the adjacencies given by the opposite criterion, obtaining $(G', A \cup A_o)$, it will have 5 adjacencies for every face point (and 6 for edge and vertex points); since they lack the interior points, the faces lose one adjacency, but they gain it back through the opposite face. The edges, which possess 4 adjacencies, gain 2, and the vertices obtain 3. Since the faces account for the largest number of points, it is clear that their limit will be 5, naturally, when scaled to large sizes.

If we recall Lemma 2.3.1, upon applying these procedures to any given board, we can always achieve the equality of $2n$ adjacencies per point, for the opposite criterion guarantees the filling of all missing adjacencies. It is easy to see that, by taking arbitrarily large boards, any n-dimensional board possesses an average of $2n$ adjacencies in the limit.

For this problem, for $n = 2$, A_{ort} provides a solution for the case with a degree as close to 4 as possible, whereas the relation $\mathcal{A} = A_{ort} \cup A_{op}$ satisfies the degree condition of exactly 4. Furthermore, for the case $n = 3$, we were able to find a partial solution on the boundary: if $G' = \{x \in G \mid \exists i \text{ such that } (x_i = 1 \vee x_i = n_i)\}$ denotes the boundary, then the graph $(G', \mathcal{A})$ is a solution for the case with a degree as close to 4 as possible on the surface.

3.2 Go in One Dimension

First, let us examine what it looks like, Figure 3.9. Clearly, there are two liberties per point, except at the boundaries, which possess one. Therefore,

$$A(n) = 2 * 1 + (n-2) * 2 = 2n - 2.$$
$$\tilde{A}(n) = \frac{2n-2}{n} = 2 - \frac{2}{n}.$$

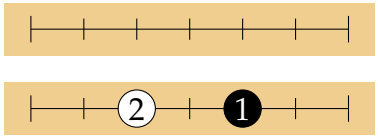


Figure 3.9 – Empty board and board with a black and a white stone.

Then, it is clear that the limit of its average is 2.

Lemma 3.2.1. *In canonical* Go^1*, the only regions that can constitute an eye are the extreme points of the board; any other vital region is a union.*

Proof. By definition, an eye for a group is a vital region completely surrounded by a single group I. Let R be a region; for I to surround R and for R to be vital, its boundary ∂R must belong to I.



Figure 3.10 – Eye at the boundary, union in the interior.

R in the interior: By Lemma 2.3.1, if $x \in R$, it possesses two adjacencies: $x-1$ and $x+1$.

For x to be an eye of I, it must hold that $\{(x-1, e), (x+1, e)\} \subset I$. Since I is connected, there would exist a path in I between $x-1$ and $x+1$. However, the geometry of the board forces the path to pass through x, which is empty; thus, such a path does not exist.

Consequently, $x-1$ and $x+1$ are disconnected and cannot belong to the same group I. An interior region is always surrounded by two distinct groups. At most, it may happen that ∂R is given by two groups of the same color, in which case it is a union.

R at the boundary: If $x = 1$, its only adjacency is point 2. If a group I possesses a stone at 2, then the entire boundary of x belongs to I. Therefore, x is vital for I and, since there is a single group surrounding x, x is an eye for I. The case $x = n$ is analogous. ●

3.3 Common Origin

Thus far, we have constructed two-dimensional Go and explored its generalizations to one and three dimensions. Although their dynamics differ, a natural question arises: are they truly distinct structures? We shall see that all of them can be derived from a single common origin.

A Go game is determined by a structure (G, E, A, Δ). To relate different variants, we shall introduce certain natural mappings that allow us to construct some variants from others while preserving the adjacencies and the rules of the game.

$$\phi : (G, E, A, \Delta) \to (G \times \{1\}, E, A, \Delta),$$

given by $\phi(g) = (g, 1)$.The preservation of adjacency is immediate; furthermore, ϕ is injective:

$$\phi(g_1) \sim \phi(g_2) \iff (g_1, 1) \sim (g_2, 1).$$

To expand the board with a set M, we make use of the mapping

$$\Xi : (G, E, A, \Delta) \to (G \times M, E, A, \Delta).$$

Fixing a coordinate $m \in M$, we define $\Xi(g) = (g, m)$.

Once again, adjacency is preserved and the mapping is injective, since $(g_1, m) = (g_2, m) \implies g_1 = g_2$:

$$\Xi(g_1) \sim \Xi(g_2) \iff (g_1, m) \sim (g_2, m).$$

To incorporate additional players, we define an inclusion

$$\iota : (G, E, A, \Delta) \to (G, E \cup \{y\}, A, \Delta),$$

with $y \notin E$, such that $\iota(e) = e$ for all $e \in E$.

It is injective ($\iota(e_1) = \iota(e_2) \Rightarrow e_1 = e_2$) and preserves the identity of previous states. By design, it is non-surjective, so as to accommodate the new element.

Let us consider $\mathrm{Go}_0^0 = (\{1\}, \{0\}, A, \Delta)$, a single-node game with no players. Knowing that $\{1\} \times N \cong_A N$, we obtain the following diagram (Figure 3.11).

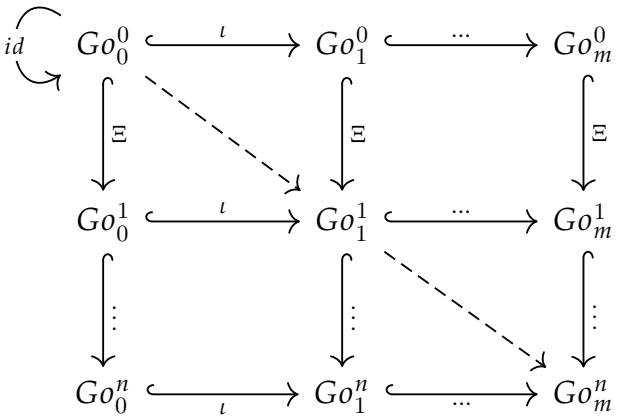


Figure 3.11 – Structure of the family of *Go Games*. Players are added to the right; the board is expanded downward.

Recalling 老子—*Laozi*, we rediscover what has already been stated:

道生一, 一生二, 二生三, 三生万物。	*The way brings forth one,* *One brings forth two,* *Two brings forth three,* *Three brings forth all things.*

道德經—DÀODÉ JĪNG CHAPTER 42

We can go a bit further; recalling what was seen in Section 3.1.4, we can define a natural transformation that associates each flat board with its toroidal version.

This mapping alters neither the number of nodes nor the players; it merely modifies the adjacency relations, identifying opposite positions on the board.

We thus define the toroidal mapping

$$\tau : (G, E, A, \Delta) \to (G, E, A \cup A_o, \Delta),$$

where A is the original adjacency relation and A_o is the relation between opposite positions.

The mapping acts as the identity on the nodes. Furthermore,

$$g_1 \sim_A g_2 \implies \tau(g_1) \sim_{A \cup A_o} \tau(g_2).$$

Since $A \subseteq A \cup A_o$, this property is immediate. In particular, τ is injective.

On the other hand, recalling the adjacency analysis carried out in Section 3.1, we observe that boundary positions exhibit a different behavior from that of interior positions.

To isolate this phenomenon, we introduce the boundary mapping

$$\partial : (G, E, A, \Delta) \to (\partial G, E, A, \Delta),$$

where $\partial G = \{x \in G : \exists i, x_i = 1 \text{ or } x_i = n_i\}$, as we defined it in Section 3.1.4.

The Hollow Torus

Combining both constructions allows us to build the diagram of Figure 3.12.

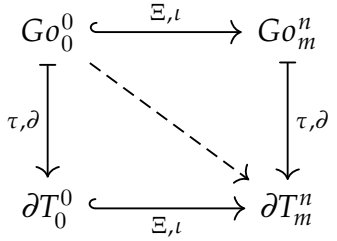


Figure 3.12 – Diagram of the complete structure. The horizontal arrows (Ξ) expand dimensions and players (ι). The vertical ones (τ) close the space, and (∂) hollow out the interior.

The Projection onto the Graph

To study the geometric properties of the board, as in Section 3.1, it is useful to temporarily ignore the state of the positions and the game dynamics. We thus define the mapping

$$U : (G, E, A, \Delta) \to (G, A).$$

This mapping associates each game with its underlying graph. In this way, properties such as the number of liberties $A(n, m)$ can be studied solely from the adjacency structure.Note that different games may determine the same graph, so the mapping is not injective.

3.4 Toward Surrounding Games

Let us see how our theory could take us further than we initially thought.

Othello

In Figure 3.13, on the Othello board, play takes place inside the squares; this poses no difficulty, as it suffices to think of the intersections as the squares. Unlike Go, stones are adjacent in all directions (including diagonals); we simply need to modify A. Furthermore, in Othello, "capturing" consists of flipping the stones to change their color. In turn, to "surround," it is only necessary to do so in one dimension—that is, to enclose a group of another color along a straight line, either orthogonally or diagonally. We can already accomplish this: it suffices to consider that upon capturing, instead of removing the stones (i.e., assigning $C(i) = 0$), we assign the color of the capturing player (by flipping them).

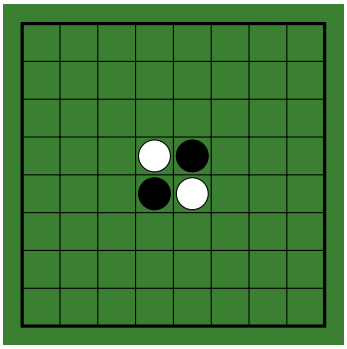


Figure 3.13 – Othello board with its initial position.

If we had Δ_{oth}, Othello can be constructed over the game: $(N \times N, \{0, 1, 2\}, A_{ort} \cup A_{diag}, \Delta_{Oth})$. With $|N| = 8$, A_{diag} being the criterion that takes diagonals $(\pm 1, \pm 1)$ into account, and Δ_{Oth} the rules of change in Othello. Thus, the developed construction suggests a possible formalization of *Othello* within the same theoretical framework.

Havannah

Another interesting example is Havannah. Unlike Go, this game takes place on a hexagonal board, and stones are adjacent according to the intrinsic geometry of said board. Once again, it suffices to modify the adjacency criterion. The particularity of Havannah is that there are three distinct ways to win: constructing a ring, a bridge, or a fork (see Figure 3.14).

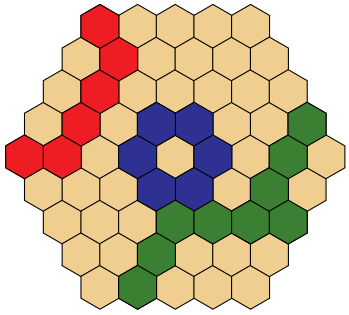


Figure 3.14 – The three winning structures in Havannah: ring, bridge, and fork.

Our theory already accounts for the case of the ring. Indeed, a ring is simply a group that encloses a region. If we distinguish the six corners of the board as special subsets of positions, a bridge can be described as a group containing stones in at least two distinct corners. Likewise, if we

distinguish the six edges of the board, a fork can be described as a group containing stones in at least three distinct edges.

Recalling the properties of $|K(i)|$ discussed in Part I, this suffices for us to identify which points belong to the corners or edges (Section 2.1). In this manner, Havannah's winning condition can be expressed solely in terms of groups, regions, and board subsets. Consequently, the game appears to admit a natural formalization within the same theoretical framework.

Hex

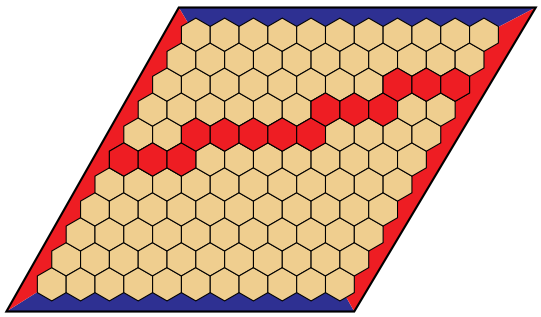


Figure 3.15 – Winning chain connecting two opposite edges.

Let us now consider Hex, Figure 3.15. The board is hexagonal; it suffices to adapt the adjacency criterion. Unlike Go, there are no captures in Hex. To win, one must construct a path from one corner to its opposite. By distinguishing the opposite edges of the board as position subsets, the winning condition can be expressed very simply: a player wins if there exists a group of their color that intersects both opposite edges of their color. Thus, victory in Hex reduces to a property of the groups.

Shannon's Game

Shannon's game constitutes a particularly interesting example; unlike Go, Hex, or Havannah, it does not take place on a board, but rather on an arbitrary finite graph. The two players alternate turns: player Short takes an edge of the graph and colors it, while Cut takes an edge of the graph and removes it. Short wins if their chosen edges contain a path connecting the distinguished vertices A and B. Cut wins if they succeed in disconnecting vertices A and B.

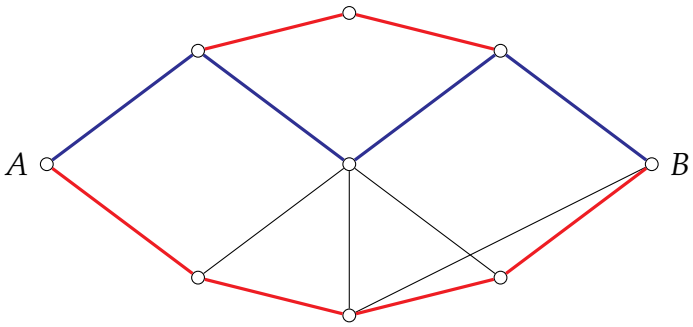


Figure 3.16 – The blue edges belong to Short, the red ones to Cut, and the black ones remain free. The set of blue edges determines a path between the distinguished vertices A and B, so Short wins.

Let $G = (V, E)$ be a finite graph, and let $A, B \in V$ be two distinguished vertices. We shall consider each edge of the graph as a playable position. Consequently, the set of

positions will be given by $T = E$. The possible states of a position will be $E_s = \{0, 1, 2\}$, where 0 represents a free edge, 1 an edge controlled by Short, and 2 an edge controlled by Cut. We define an adjacency relation $\sim_s$ on T: two positions $e_1, e_2 \in T$ are adjacent when the corresponding edges share a vertex—that is,

$$e_1 \sim e_2 \iff e_1 \cap e_2 \neq \emptyset.$$

With this definition, the groups introduced earlier coincide with connected components of edges of the same color. We denote the game dynamics by Δ_s; any Shannon game associated with the graph G can be represented by the structure:

$$(T, E_s, \sim_s, \Delta_s).$$

This example is especially revealing, as it shows that the developed structure does not depend on geometric notions such as regions or territories. Even though it is a game defined on an arbitrary graph, the notions of position, state, adjacency, and group remain sufficient to completely describe its dynamics. It appears that the developed theory describes not only Go, but a broader family of games: *Surrounding Games* (SG).[2]

[2]This name is not entirely accurate, but it is the one that emerged during development. It is adopted here for convenience and does not pretend to identify an established branch of mathematics. Determining an appropriate designation would require a broader study.

3.5 Conclusion

In this work, we have constructed a theory capable of formally describing classical Go from elementary notions—*where, who, what, and how*—supported by set theory. Throughout this construction, the rules of the game emerge naturally, and intuitive concepts acquire a precise and tractable expression.

However, the principal value of this framework is not limited to formalizing Go. As explored in Part III, the developed structure is sufficiently general to encompass a broader family of games based on the idea of surrounding and capturing. This suggests the existence of a more abstract mathematical object, which we designate as SURROUNDING GAMES (SG), of which Go is merely a particular case.

Several paths remain open: identifying which properties are exclusive to Go and which belong to the entire family, formalizing this family and seeing how far we can take it, constructing infinite Go, or attempting to connect this theory with combinatorial game theory.

I'm actively thinking about this, I don't have my whole time for this but I will, from time to time make updates with new stuff (and fixes). Right now I'm most intereted is in attacking the infinte Go. The biggest problems are, as far as I can understand, how to define an infinte capture and SuperKo detection.

IV: Infinite Go

This chapter it's a draft, at least for now. I'll include my thought process and most of what I'm able to understand/formalize about it.

To begin with, let's see *where* we are playing now. Our infinite board should be able to have also infinite dimensions and every one of those could or not be infinite as well.

We'll begin with the simplest case, let's play on a two dimensional board that has one infinite dimension.

Let Γ be a Go game such that,

$$G = \{1, 2, \ldots, n\} \times \mathbb{Z}.$$

for $n \in \mathbb{N}$. We'll use $\mathbb{Z}$ for simplicity as it's a better (easier) representation of the infinite board. As the number of players does not matter at all, let's use only two players and, the canonical adjacency. Our game will be $\Gamma = (G, E, A_{ort}, \Delta_{Go_\infty})$. At least for now, we are not sure how Δ_{Go_∞} works but, we will get there.

Our board could (and for us, will) look like this, Figure 4.1. Now, there is a substantial change in this new game.

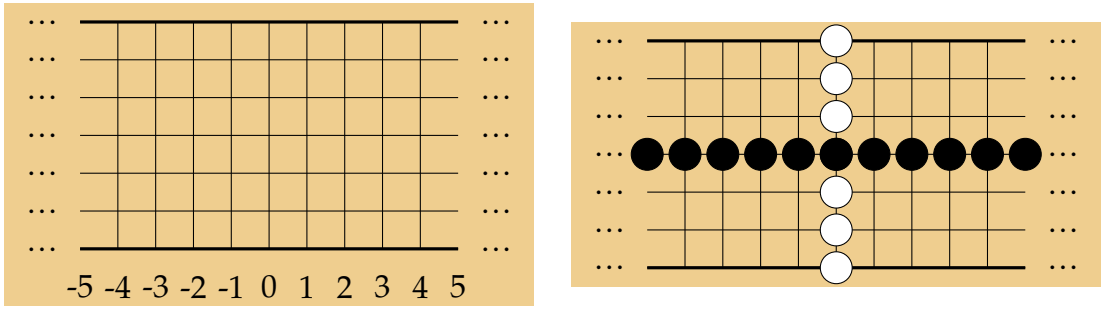


Figure 4.1 – A $7 \times \infty$ board and two intersecting plays that resolves by definition.

Notice that, if the white line were infinite, such play generated two infinite lines in one turn, which is crazy.

The very first problem on the infinite board it's how to determine a capture. Any other prior definition should work just fine.

Lets say we are in the next position, Figure 4.2

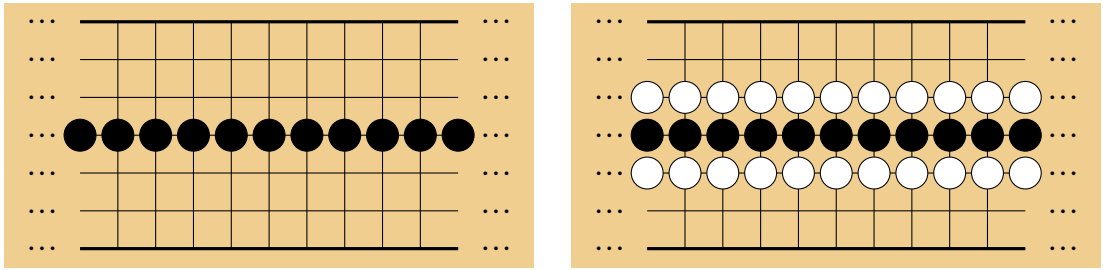


Figure 4.2 – An infinite board with an infinite black group being capture.

As this group is infinite, it will touch the very end of the board so we can capture it by surronding it at the top and bottom.

We can be sure that this is capture because if the black group it's I, then $K(I) \subset S$ where S is the set of the two white groups. That is, $L(I)$ should be 0. Now let's take a look at another example, an infinite group but only on one side.

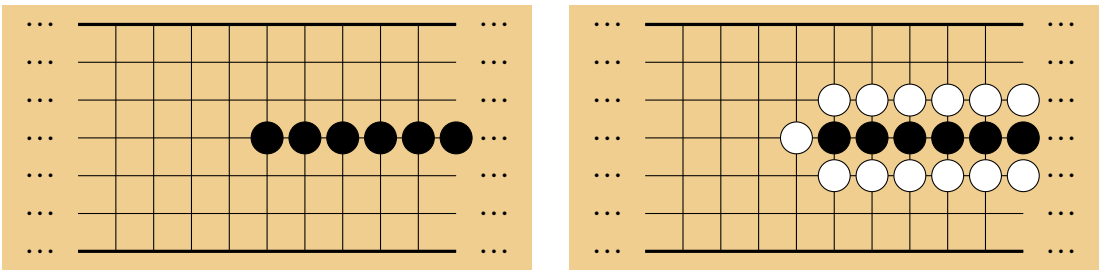


Figure 4.3 – An infinite, on one side, black group being capture.

In this case, not using the whole board, even though it's infinite, its better. As it provides one more liberty (or does not loses it).

The following example reflects and interesting nature of inifinite Go. (More on this later)

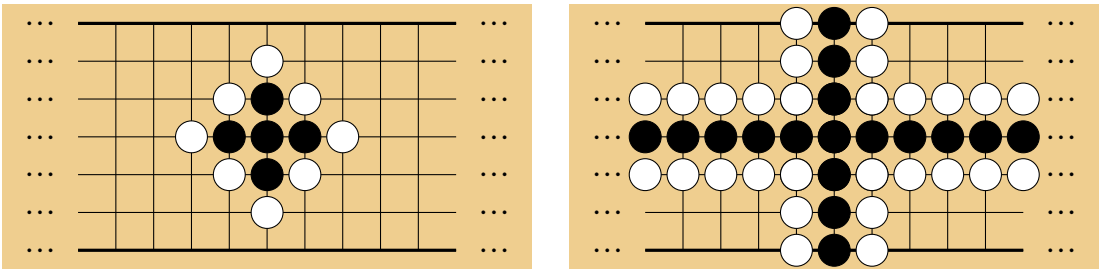


Figure 4.4 – Two captures.

It seems that finite groups are stronger than infinite ones as, again, they don't lose liberties due to not being infinite.

Lets characterize *what* can be done in the game.

Definition 4.0.1 (A move)**.** In an infinite game of Go of dimension n, a player in their turn t can perform one of the following actions:

1. Pass.

2. Propose a connected hyperplane H of dimension at most $n - 1$. The hyperplane H is a subset of the board such that, for at least one dimension, all its points have a fixed coordinate. All the stones in H have the color of the player making the move. In particular, if $\dim H = 0$, then H is asingle stone.

   The stones already present on the board may intersect H. Let

   $$P = \{P(x) \in T_t : C(x) \neq 0\}$$

   be the set of occupied points at turn t. The stones that are actually played are then

   $$H - P = \{h \in H : P(h) \notin P\}.$$

The set $H \setminus P$ may consist of several groups, as the stones already present on the board may disconnect the proposed hyperplane.

If any group contained in $H - P$ is in suicide condition, the proposed move is rejected. Otherwise, the stones in $H - P$ are placed on the board, and captures are resolved according to the usual rules.

Let me explain why $n - 1$ dimensions.

If we look back at the game of Go, stones are placed on a two dimensional board. In our formulation, this means that we place a connected 0 dimensional subset of the board, that is, a single stone. In other words, the dimension of the object being played is $n - 2$ relative to the dimension of the board.

If we kept this relation, we would obtain a natural generalization for higher dimensional boards, but we wouldn't be unable to define a move in one dimensional Go, since this would require

$$n - 2 = -1.$$

Thus, in an n dimensional game, a player may propose a connected hyperplane H satisfying

$$0 \leq \dim H \leq n - 1.$$

4.1 Infinite SuperKo

We cannot use Zobrist method for the infinite Go as we will need infinite many bits. We need a better method, let's try to make one.

If we remember the nature of the SuperKo rule, it's says that we cannot replicate a prior board via a stone play. Again, we are using this version of SuperKo (Positional).

Because of the rule, every game has to be a line, that is, as we cannot go back into the history of plays (at least it's a pass but will not take in consideration those). From the empty board to the very end, we have a straight forward path. Not even that but, because of the rule, the whole game of Go its a *tree*. That is, starting from the empty board, there are 362 possbile plays (361 stone placements and a pass) and we cannot go back. From those, every posible outcome, that is, any end of a match can be found by following a line of unrepeated boards.

Let's define a way to measure how close are two given boards are.

First, lets notice that, in the very beggining, there are exactly one region and not much else in the board. That is, before the first turn, $|J| = 1$ (the big empty region).

Now, the very first turn will make so $|J| = 1$ if the black player plays a stone or a finite line. If the player chooses to play an infinite one, then $|J| = 2$. Lets pretend it just played an infinite group, with that, Zobrist method or any method that uses the stone itself go nuts and can't be use here but, $|J|$ it just a little bigger, finitely bigger. Now, this is not perfect as there are infinite many ways to play an infinite group so we end up with $|J| = 2$. Before going forward, lets see an example.

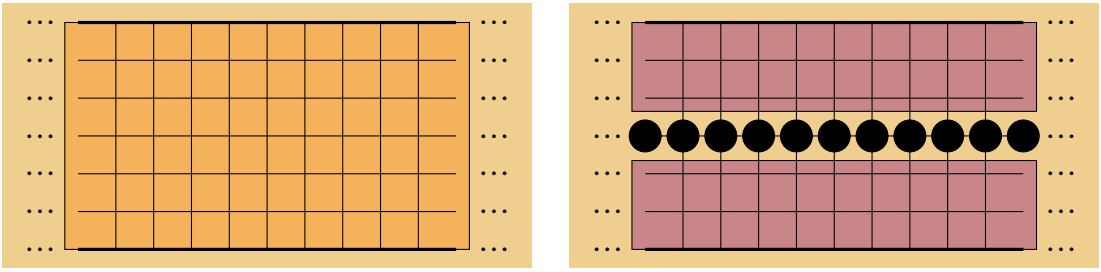


Figure 4.5 – A board where the number of regions merely grows by one when an infinite group is played.

As I said early, playing such group vertically (in this case will be finite but you get the idea) or shifting it up or down, will end up with the same number of regions.

The first problem is to determine if the number of regions are always finite or not.

The second problem is how to determine those boards that has the same number of regions are infact different, we cannot look at them as that's too costly. My idea is, if we can solve the first problem, then we can measure every region and, by that, we can tell even if two boards have the same number of regions, those regions might not be equall (by biggest and smallest radius or something like that).

The third problem its color shifting, two boards might not be equal but the regions could be exactly the same (number and size), but the colors of the regions surronding them might not. (I'm thiking about discrinating between finite and infinite regions. Then, in the set of finites, we can calculate the influence of each color, this should work but its just another filter and its far from perfect)

Lets attack the first problem but we need to be a little more precise of what we want.

Given an infinite game of Go, the number of regions are always finite? If there are infinite moves then the answer is no. Lets consider the infinite one dimensional Go.

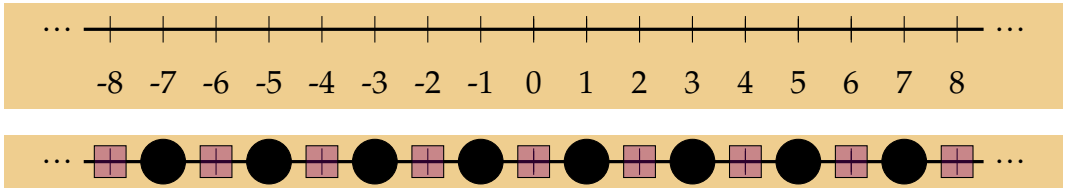


Figure 4.6 – A one dimensional infinite Go board and a example of infinite many regions by infinite many plays.

Is it possibe to do many examples like this that end up with infinite many regions. The trick here and, perhaps, my salvation, its that there are infinite plays, which is, given the nature of the game, something that shouldn't happen. If we restrict ourselves into only finite many plays, then our question becomes:

Given an infinite game of Go, the number of regions generated by a finite number of plays are always finite too? More preciselly, we want to prove that,

$$t < \infty \Rightarrow |J(T_t)| < \infty.$$

In order to do it, we need to show something about the nature of the game, we'll characterize how to split a region. For that, we need some tools first.

Definition 4.1.1 (Dense hyperplane)**.** Given a proposed hyperplane H, we say that H is dense if every stone $t \in H$ is occupied, that is,

$$C(t) \neq 0 \quad \forall t \in H.$$

Lemma 4.1.1 (Adjacency bound)**.** *Given a $n \leq \infty$ dimensional board of Go, with the canonical adjacency, a point t in the board, has at most $2n$ adjacencies and at least n.*

$$n \leq |t| \leq 2n.$$

Proof. This is left as an exercise for the reader. ●

Theorem 4.1.2. *In an n-dimensional game of Go, a move that splits a region R of dimension d must be a dense hyperplane of dimension at least $d - 1$.*

This will tell *how* split a region but not exactly *when*. We'll get there.

Theorem 4.1.3 (Finite regions induced by finite plays)**.** *Given an infinite game of Go, the number of regions generated by a finite number of plays are always finite.*

Proof. ●

We are going to do an induction in the number of dimensions and the number of turns. if t its for turns and n its for dimensions, we'll have the following the game.

for $n = 1, t = 1$, we have at most two regions as we describe, we can only play single stones here. Then the placed stone splits the region into two.

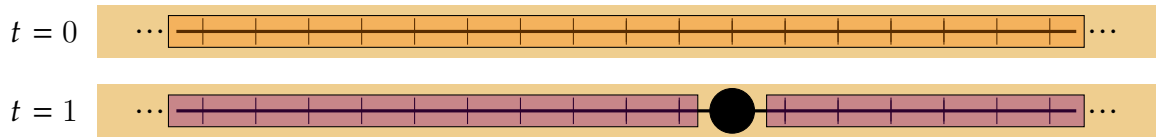


Figure 4.7 – An infinite one dimension and the first turn that shows how a region it's split.

Lets assume that for the turn $t - 1$, the number of regions its finite.

for the t turn, the proposed play, given its legal, occupies at most every region. if p its the number of regions at turn t, then we at most can splip every region, capture a finite number of groups (give we have play a finite number of turns). Therefore, te amount of regions stills finite.

now lets assume that for a $n - 1$ go board, the number of regions stills finite.

$n = n, t = 1$.

Again, the board begins empty and there is exactly one region, by playing whatever hyperplane, the number of regions stills finite.

Lets assume that, for n dimensions and $t - 1$ turns, the number of regions stills finite.

For the t turn, we can only play in such way we touch at most a finite number of regions, the ones on turn $t - 1$. Lets suppose we can interact with every region. The result of such play will end up creating new regions, by capturing. Filling regions (assuming legality) or splitting them if we join an existent group. Finally the number of them stills finite.

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